

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)
```

```

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()

```

```

        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"):
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}

```

```

elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test

- `train_cv`: mean score train of the CV
- `test_cv`: mean score test of the CV
- `test`: score on the testing dataset
- `diff`: `test-test_cv`
- `r_test`: `test/test_cv`
- `r_train`: `train_cv/test_cv`

Colors:

[Orange] `r_test` < 0.95 (under performance of the test)

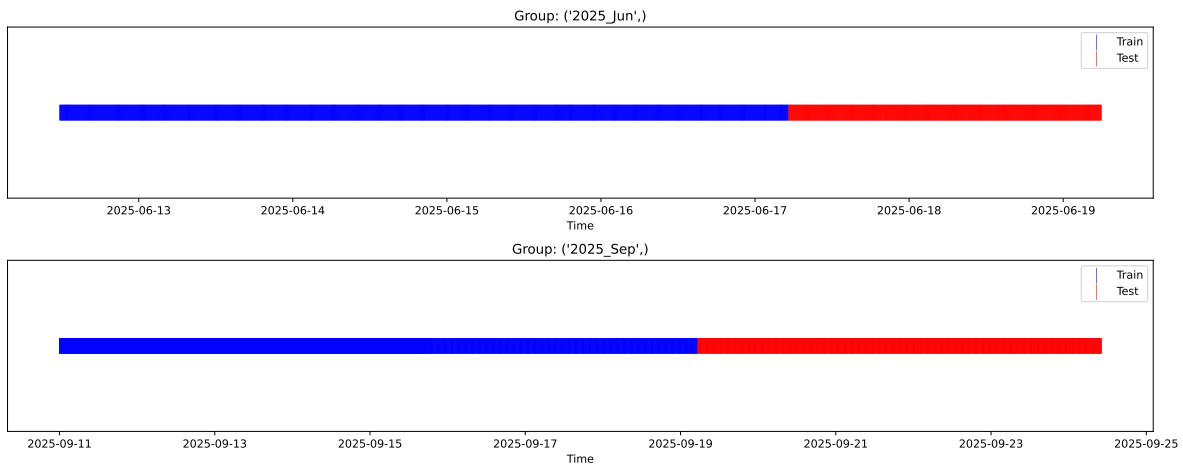
[Green] `r_test` > 1.05 (over performance of the test)

[Red] `r_train` not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: `X_train` vs `X_test`

External Split: Train: 5719 samples Test: 2449 samples



2.2.2 Internal CV Folds

Fold 1/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



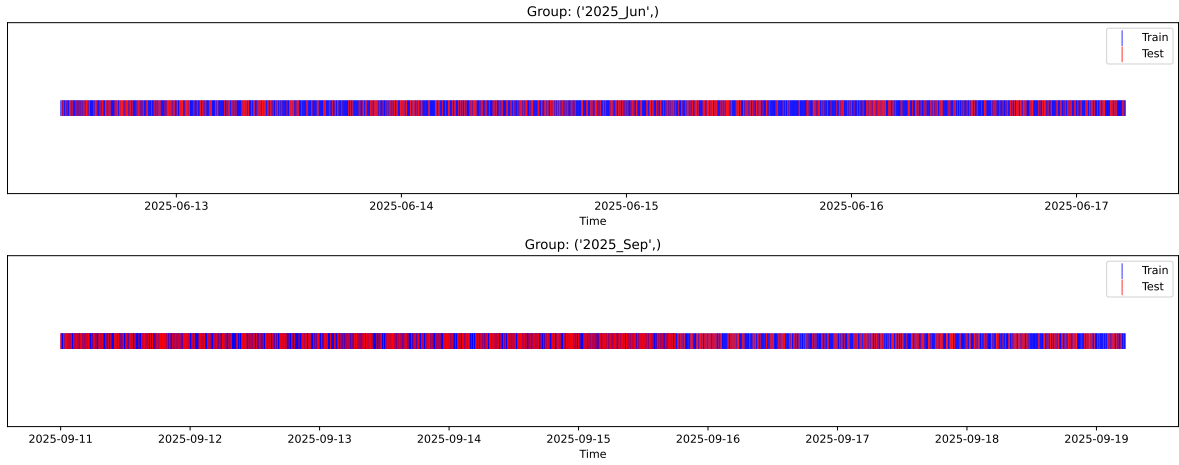
Fold 2/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



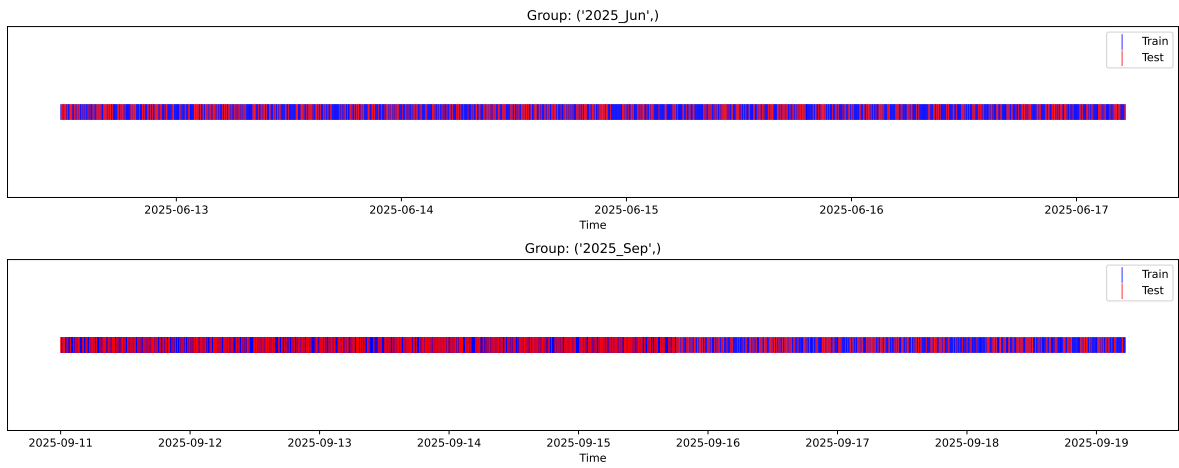
Fold 3/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



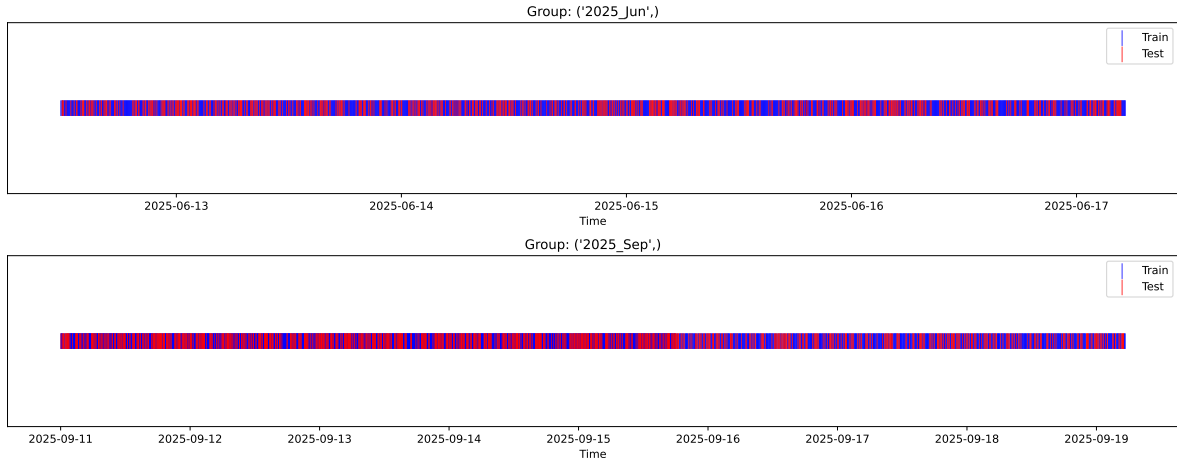
Fold 4/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.804, min=0.959, max=8.682
 y_{test} : mean=2.340, std=0.772, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 1 — y_{test} : mean=2.37, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.80, y_{train} : mean=2.35, std=0.81

Fold 3 — y_{test} : mean=2.33, std=0.82, y_{train} : mean=2.36, std=0.80

Fold 4 — y_{test} : mean=2.33, std=0.77, y_{train} : mean=2.36, std=0.82

train scores CV: [0.40125516 0.40930372 0.40130967 0.40621486 0.39718676]

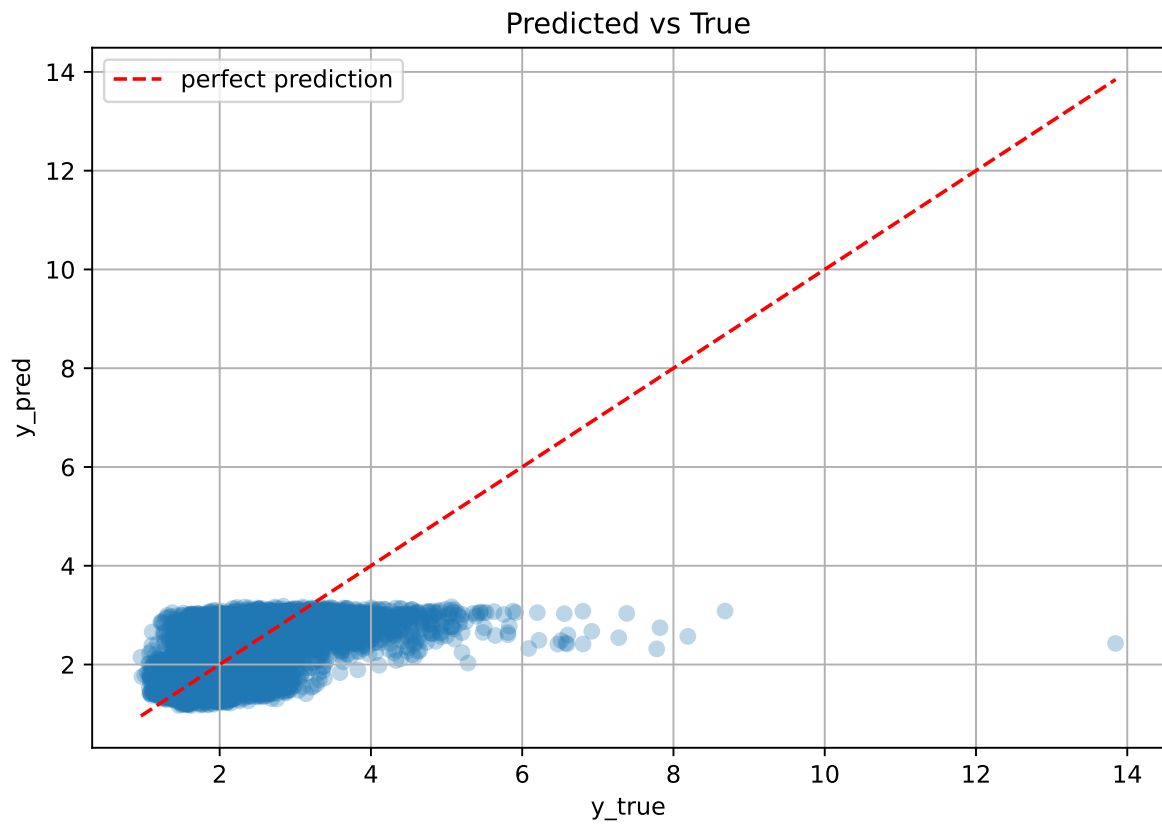
test scores CV: [0.41310399 0.39293914 0.41249368 0.39999369 0.42378683]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4031	0.4085	0.0108	0.1872	0.1866	-0.2219	0.4568	0.9868

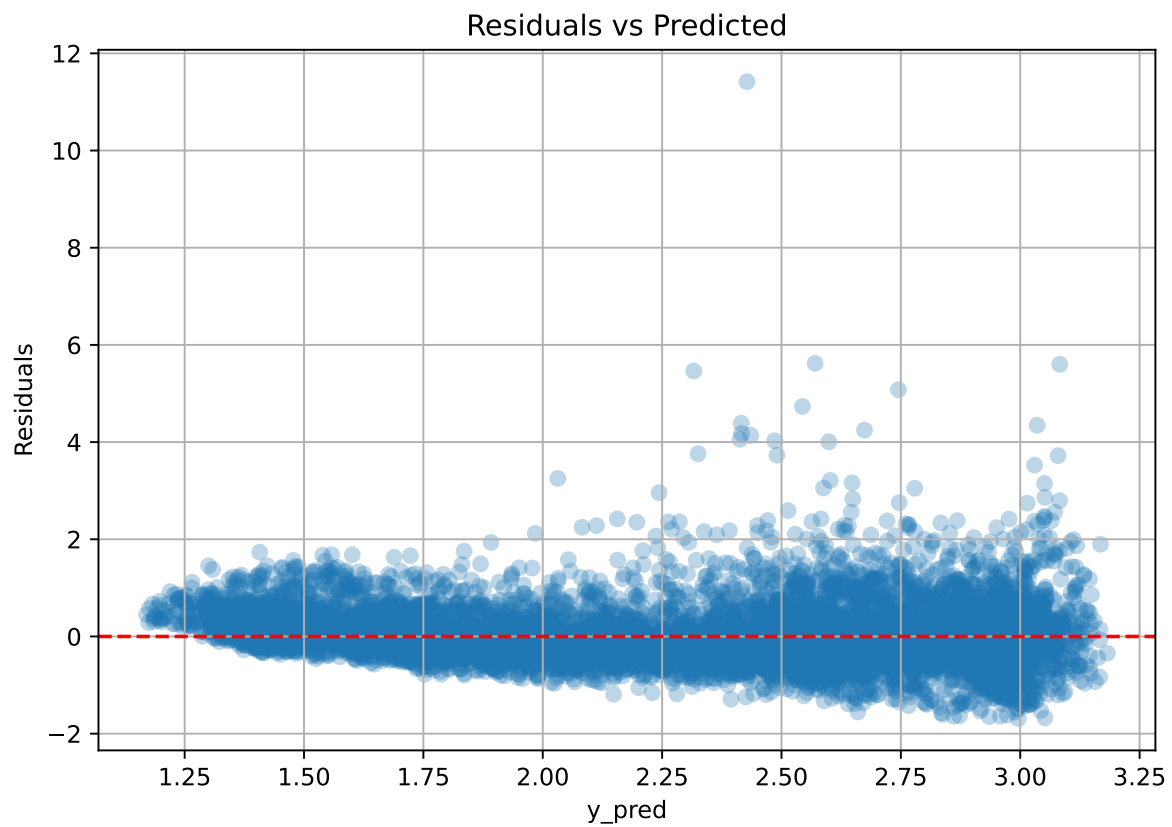
Coefficients:

coef_r_CO2 = -0.0195, coef_r_NH3 = -0.0421, coef_r_THI = 0.8947, coef_r_temperature = -1.1610, coef_r_umidity = -0.8420, intercept = 2.3535,

2.3.2 Scatter



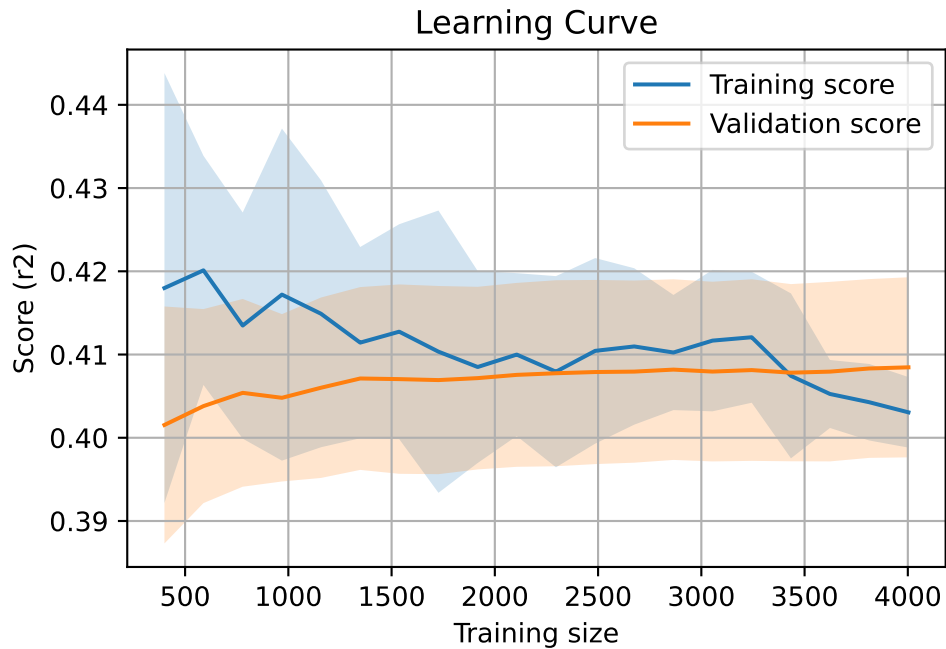
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

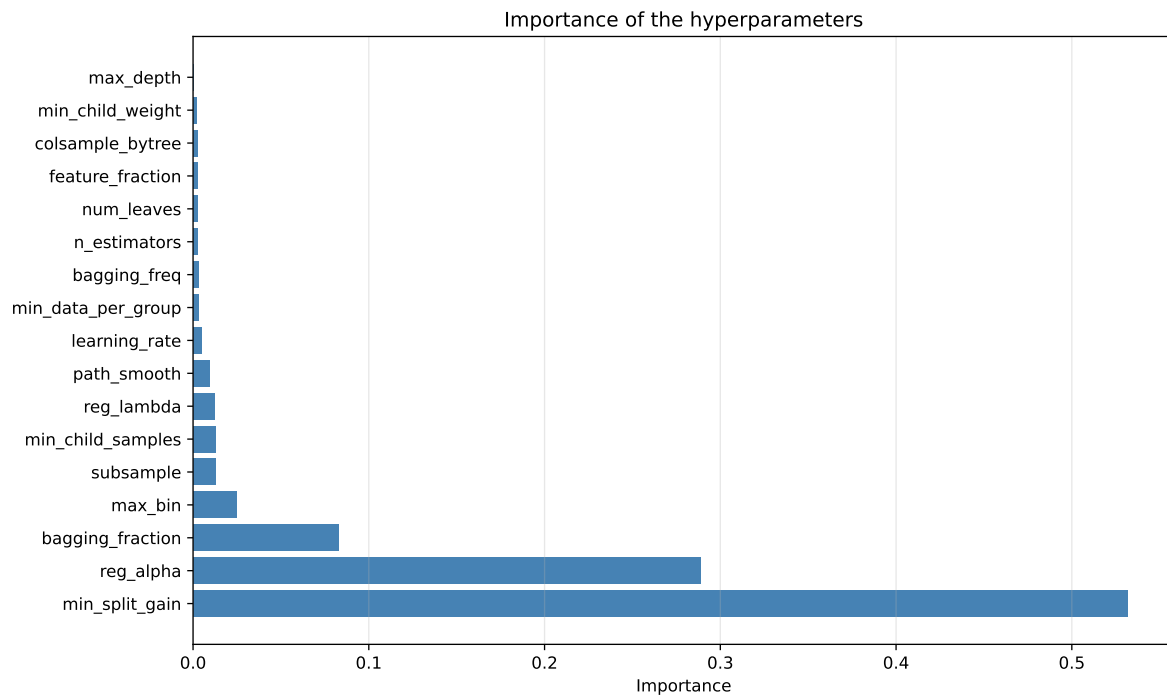
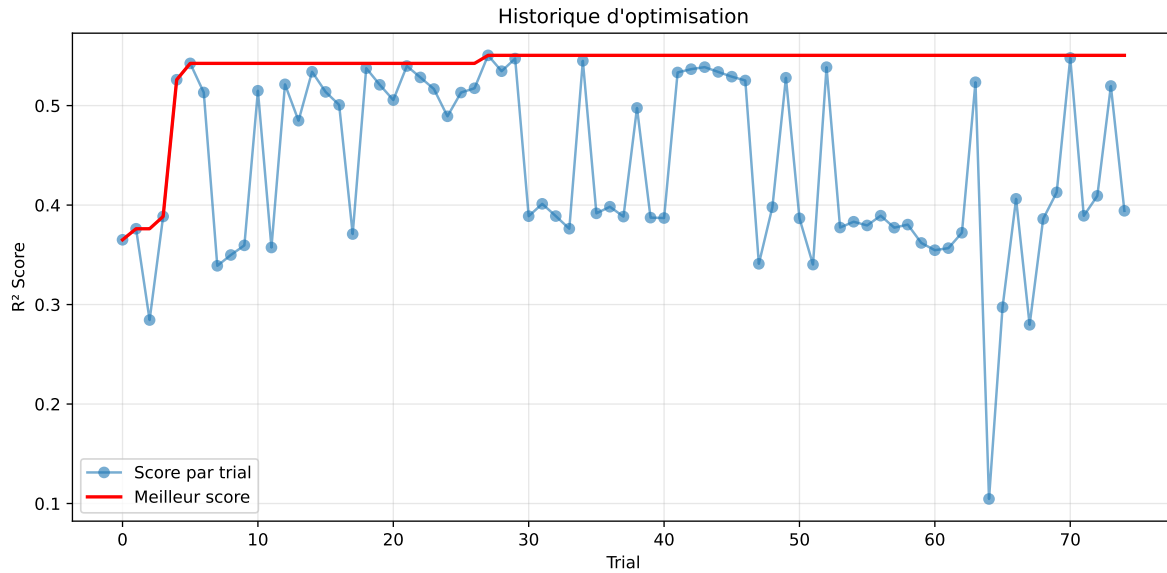


2.4 Model : LGBM

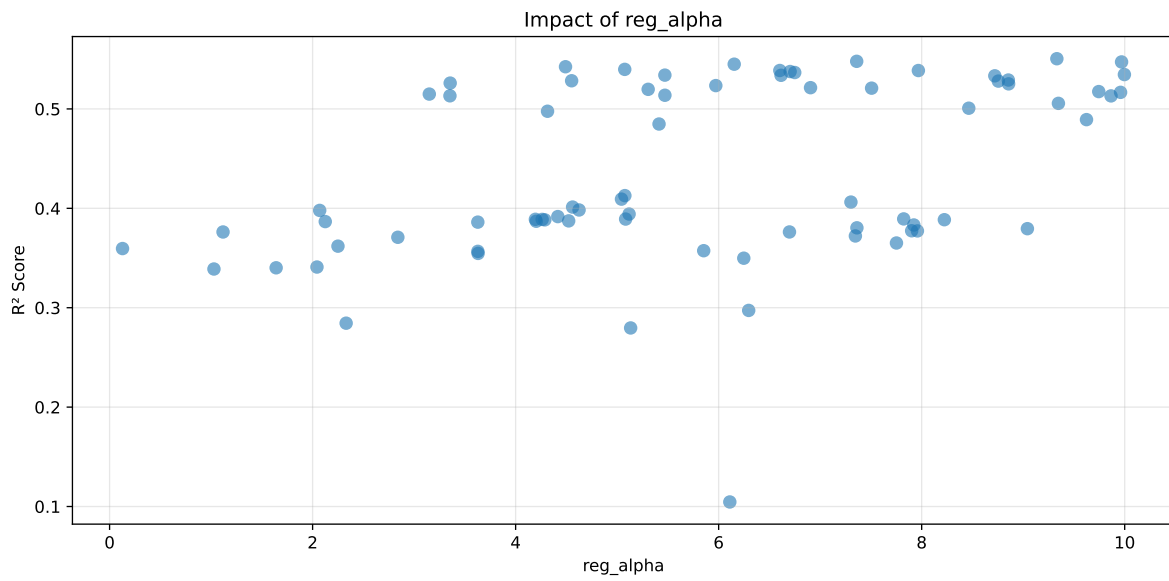
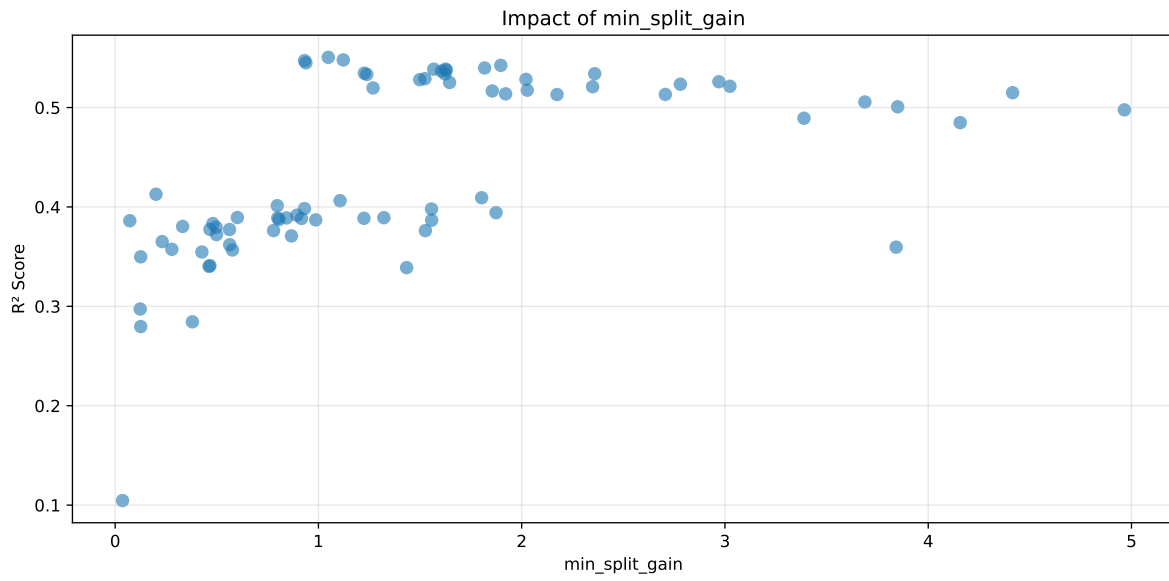
2.4.1 Gridsearch

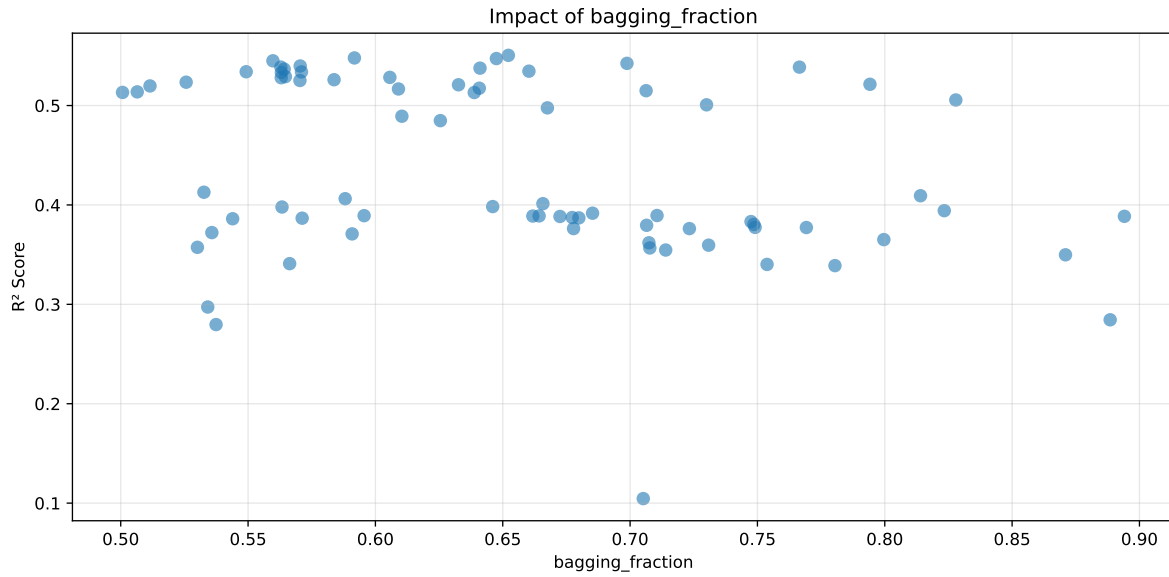
Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 1485 r_umidity 1059 r_NH3 902 r_THI 787 r_temperature 671



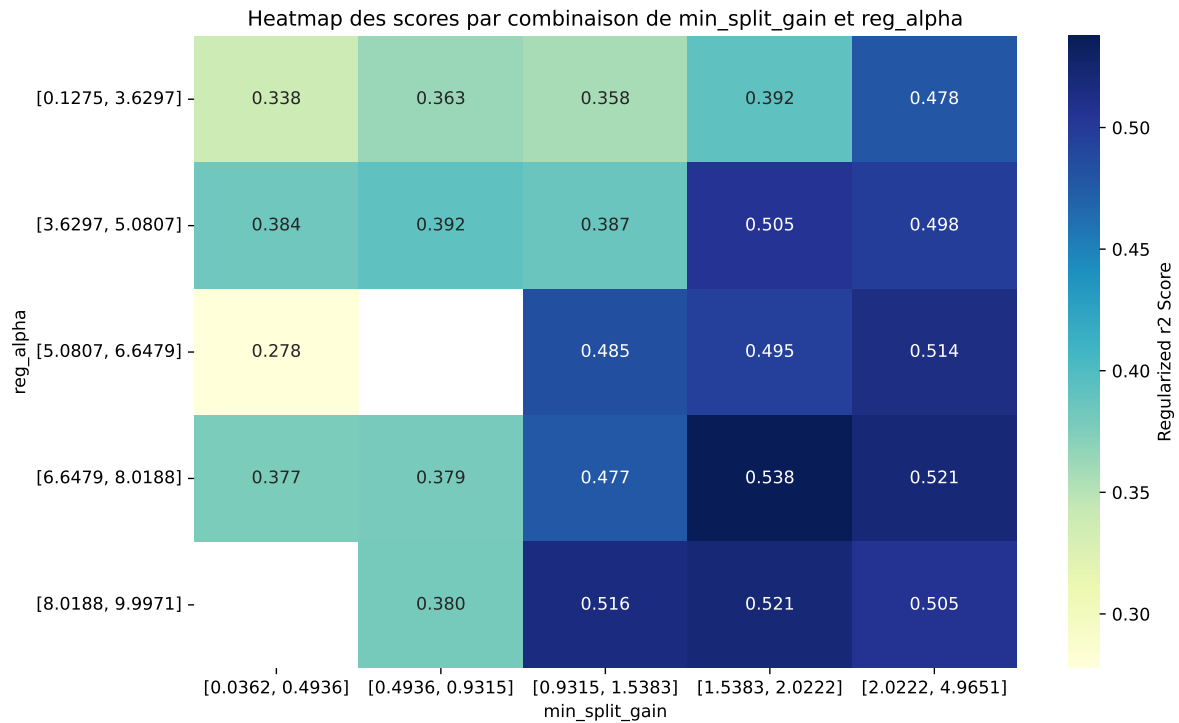
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'bagging_fraction']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.578	0.5505	0.2745	0.0191	0.5505	0.2662	-0.2843	0.4836	1.0499
2	0.5741	0.5479	0.2754	0.0178	0.5479	0.271	-0.2769	0.4946	1.0479
3	0.5696	0.5473	0.2739	0.0185	0.5473	0.2615	-0.2858	0.4777	1.0409
4	0.5701	0.545	0.277	0.017	0.545	0.2732	-0.2718	0.5013	1.046
5	0.5671	0.5424	0.2776	0.0195	0.5424	0.2741	-0.2684	0.5053	1.0455
6	0.5647	0.5398	0.2807	0.017	0.5398	0.2754	-0.2644	0.5102	1.0461
7	0.5612	0.5387	0.2741	0.0167	0.5387	0.2687	-0.27	0.4988	1.0419
8	0.5651	0.5386	0.2755	0.0163	0.5386	0.2753	-0.2634	0.511	1.0492
9	0.5621	0.5376	0.2731	0.023	0.5376	0.2675	-0.2701	0.4976	1.0455
10	0.5605	0.5366	0.2751	0.0187	0.5366	0.2698	-0.2668	0.5028	1.0445
---	---	---	---	---	---	---	---	---	---
75	0.7325	0.6278	0.2193	0.0204	0.1046	0.2073	-0.4206	0.3301	1.1667

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.010570309524016228, 'reg__num_leaves': 17, 'reg__subsample': 0.8762091484281441, 'reg__colsample_bytree': 0.7980554266036767, 'reg__min_child_samples': 25, 'reg__reg_alpha': 9.33173364979183, 'reg__reg_lambda': 48.8176237933237, 'reg__min_split_gain': 1.0483440338500922, 'reg__path_smooth': 1.336311309623837, 'reg__min_child_weight': 9.537612469969497, 'reg__feature_fraction': 0.8641233996613971, 'reg__bagging_fraction': 0.6522052331803719, 'reg__bagging_freq': 1, 'reg__max_bin': 81, 'reg__min_data_per_group': 150}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.012565802846093889, 'reg__num_leaves': 31, 'reg__subsample': 0.8765547692801698, 'reg__colsample_bytree': 0.7549511901167433, 'reg__min_child_samples': 15, 'reg__reg_alpha': 7.359852375688995, 'reg__reg_lambda': 30.724639448829304, 'reg__min_split_gain': 1.1223146972294762, 'reg__path_smooth': 4.26381425072502, 'reg__min_child_weight': 3.9923321443707516, 'reg__feature_fraction': 0.8418835575291113, 'reg__bagging_fraction': 0.5917601171044133, 'reg__bagging_freq': 2, 'reg__max_bin': 101, 'reg__min_data_per_group': 157}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.02934116116869873, 'reg__num_leaves': 18, 'reg__subsample': 0.7763214221541828, 'reg__colsample_bytree': 0.8020845716222662, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.96917707365961, 'reg__reg_lambda': 48.96084502143949, 'reg__min_split_gain': 0.931675969916147, 'reg__path_smooth': 1.35255971640954, 'reg__min_child_weight': 6.710521045255128, 'reg__feature_fraction': 0.8731304185004547, 'reg__bagging_fraction': 0.6475014645254198, 'reg__bagging_freq': 5, 'reg__max_bin': 71, 'reg__min_data_per_group': 151}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.01423147942811168, 'reg__num_leaves': 43, 'reg__subsample': 0.77628784319059, 'reg__colsample_bytree': 0.8316578270047066, 'reg__min_child_samples': 30, 'reg__reg_alpha': 6.153264585843402, 'reg__reg_lambda': 41.63914651613792, 'reg__min_split_gain': 0.9383520454244124, 'reg__path_smooth': 0.006540885669252372, 'reg__min_child_weight': 1.5392014459913128, 'reg__feature_fraction': 0.8310808603713229, 'reg__bagging_fraction': 0.5597603257578979, 'reg__bagging_freq': 2, 'reg__max_bin': 210, 'reg__min_data_per_group': 200}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.032979501528918036, 'reg__num_leaves': 24, 'reg__subsample': 0.8017988715018783, 'reg__colsample_bytree': 0.8471225701974563, 'reg__min_child_samples': 48, 'reg__reg_alpha': 4.491382780198042, 'reg__reg_lambda': 49.03578089254118, 'reg__min_split_gain': 1.8971290223969184, 'reg__path_smooth': 4.225572662684154, 'reg__min_child_weight': 0.0019186098960449708, 'reg__feature_fraction': 0.6019592183375408, 'reg__bagging_fraction': 0.6987702328497514, 'reg__bagging_freq': 3, 'reg__max_bin': 167, 'reg__min_data_per_group': 157}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.010608802817481824, 'reg__num_leaves': 60, 'reg__subsample': 0.8666919787962329, 'reg__colsample_bytree': 0.8982462676372388, 'reg__min_child_samples': 22, 'reg__reg_alpha': 5.074891180946301, 'reg__reg_lambda': 27.835062742287548, 'reg__min_split_gain': 1.8175730111108332, 'reg__path_smooth': 1.3847862428201978, 'reg__min_child_weight': 1.055620350444531, 'reg__feature_fraction': 0.8521317361571367, 'reg__bagging_fraction': 0.5704722725400198, 'reg__bagging_freq': 4, 'reg__max_bin': 199, 'reg__min_data_per_group': 144}

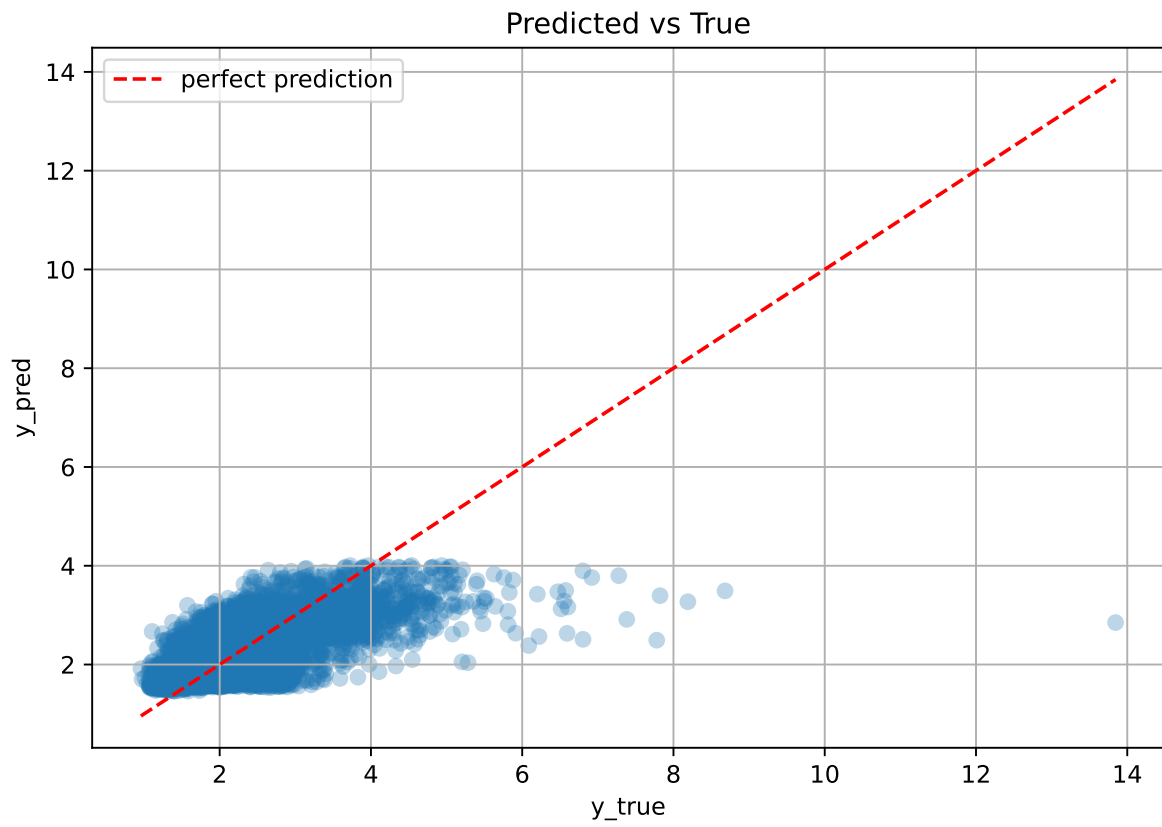
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.04357505037269124, 'reg__num_leaves': 27, 'reg__subsample': 0.8971559966757348, 'reg__colsample_bytree': 0.8312109058182862, 'reg__min_child_samples': 31, 'reg__reg_alpha': 6.6024971995352235, 'reg__reg_lambda': 34.29122373928921, 'reg__min_split_gain': 1.6253465640084017, 'reg__path_smooth': 2.2326126516832274, 'reg__min_child_weight': 2.465329129233368, 'reg__feature_fraction': 0.7945645371304547, 'reg__bagging_fraction': 0.5628838318490773, 'reg__bagging_freq': 6, 'reg__max_bin': 210, 'reg__min_data_per_group': 123}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.017756139954846498, 'reg__num_leaves': 27, 'reg__subsample': 0.7331664155960423, 'reg__colsample_bytree': 0.863353657589225, 'reg__min_child_samples': 32, 'reg__reg_alpha': 7.967578320482755, 'reg__reg_lambda': 44.76456157341222, 'reg__min_split_gain': 1.5672965968258905, 'reg__path_smooth': 0.9370207432169106, 'reg__min_child_weight': 0.49101880090119454, 'reg__feature_fraction': 0.6310748031311847, 'reg__bagging_fraction': 0.7664963356537409, 'reg__bagging_freq': 3, 'reg__max_bin': 230, 'reg__min_data_per_group': 116}

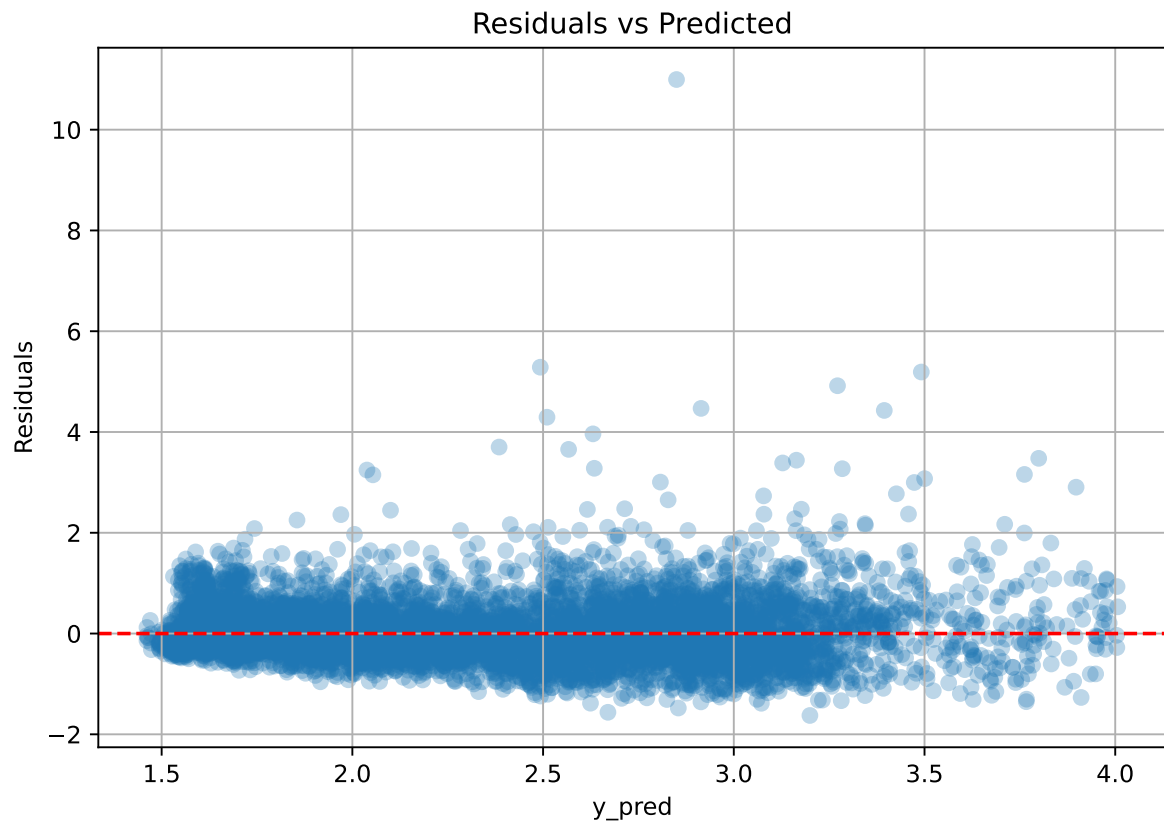
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.09359903952401369, 'reg__num_leaves': 27, 'reg__subsample': 0.7252232108829922, 'reg__colsample_bytree': 0.7899763489221121, 'reg__min_child_samples': 31, 'reg__reg_alpha': 6.704133659118151, 'reg__reg_lambda': 46.38500305448927, 'reg__min_split_gain': 1.6291655298143466, 'reg__path_smooth': 0.037103252412438725, 'reg__min_child_weight': 3.161585894730825, 'reg__feature_fraction': 0.7885560627307027, 'reg__bagging_fraction': 0.6410745059977846, 'reg__bagging_freq': 7, 'reg__max_bin': 242, 'reg__min_data_per_group': 103}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.04125603182924071, 'reg__num_leaves': 28, 'reg__subsample': 0.8984705579781438, 'reg__colsample_bytree': 0.8354344092861278, 'reg__min_child_samples': 32, 'reg__reg_alpha': 6.749503118545283, 'reg__reg_lambda': 35.86767688428295, 'reg__min_split_gain': 1.605508192451564, 'reg__path_smooth': 2.0167220233055425, 'reg__min_child_weight': 2.5861062796542016, 'reg__feature_fraction': 0.8024373109890396, 'reg__bagging_fraction': 0.5641809573097007, 'reg__bagging_freq': 6, 'reg__max_bin': 213, 'reg__min_data_per_group': 146}

2.4.2 Scatter



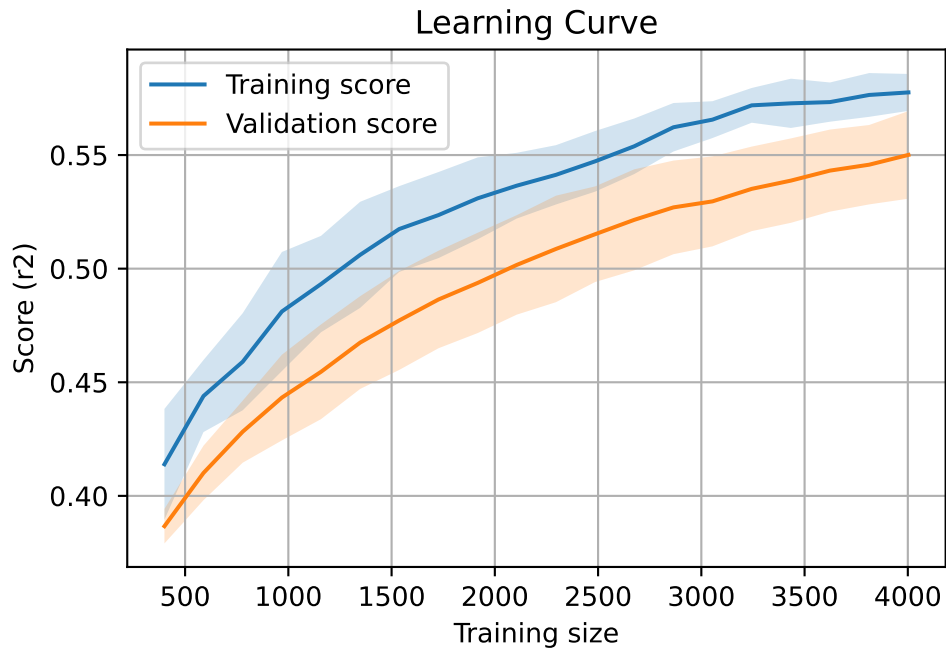
2.4.3 Scatter residuals



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

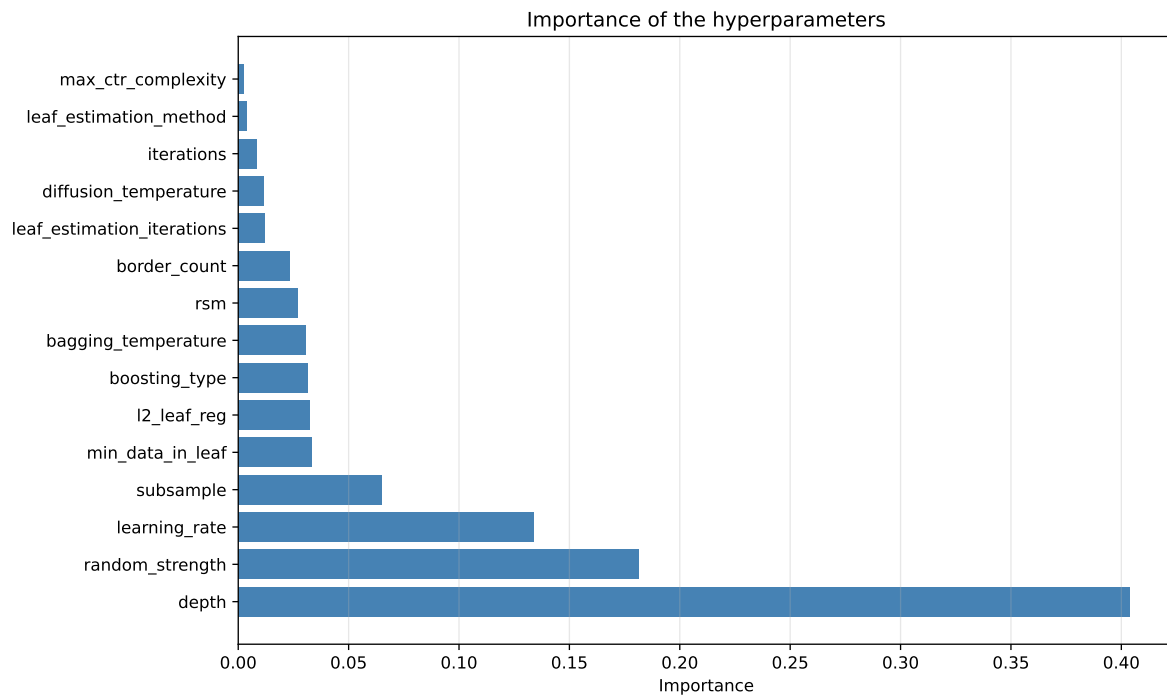
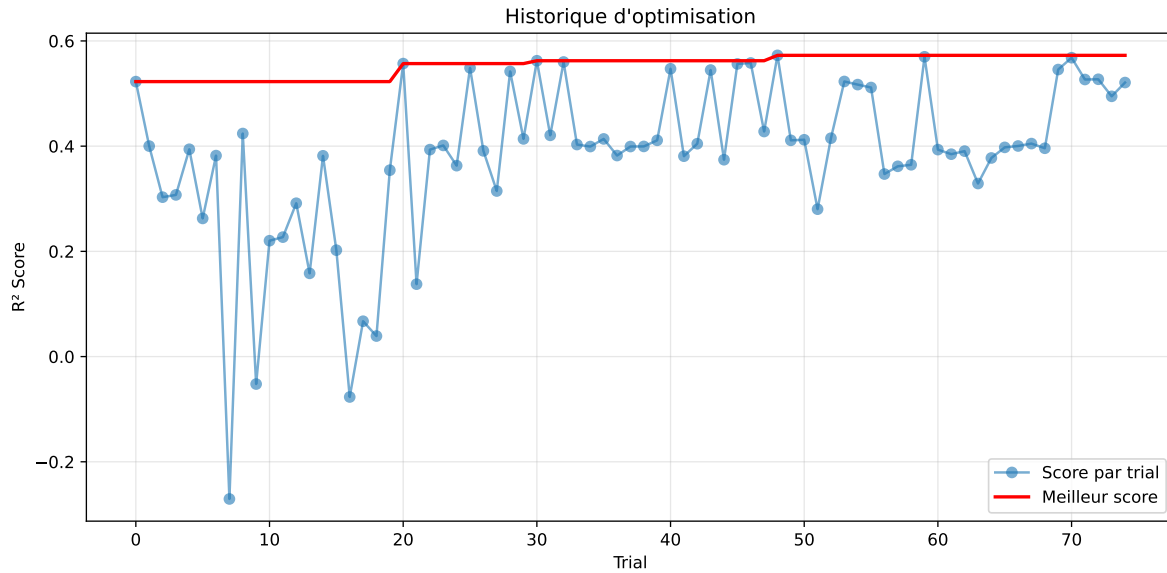


2.5 Model : CatBoost

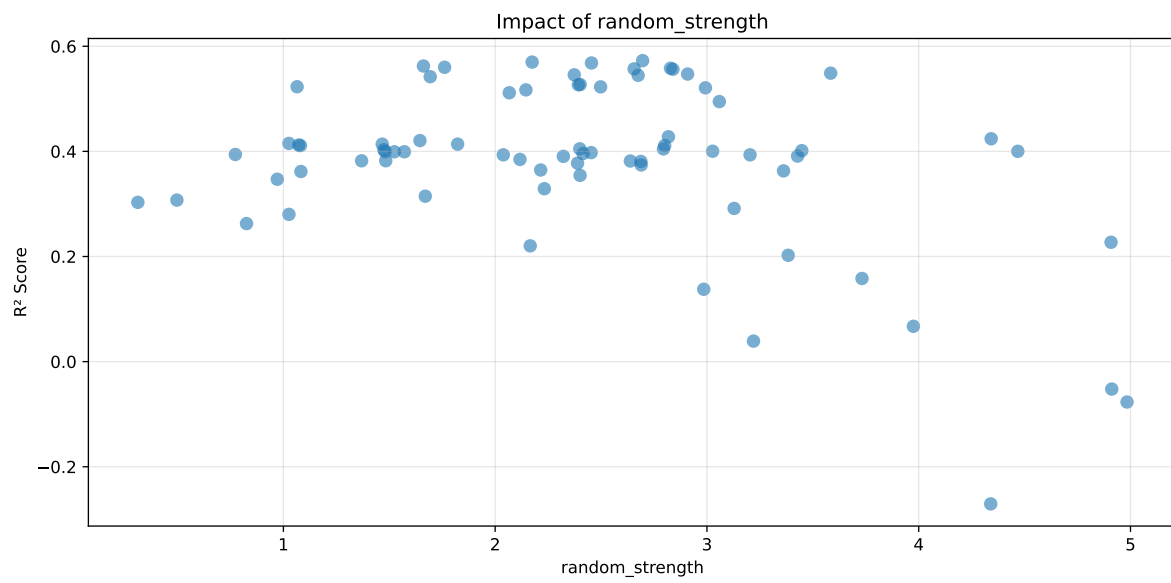
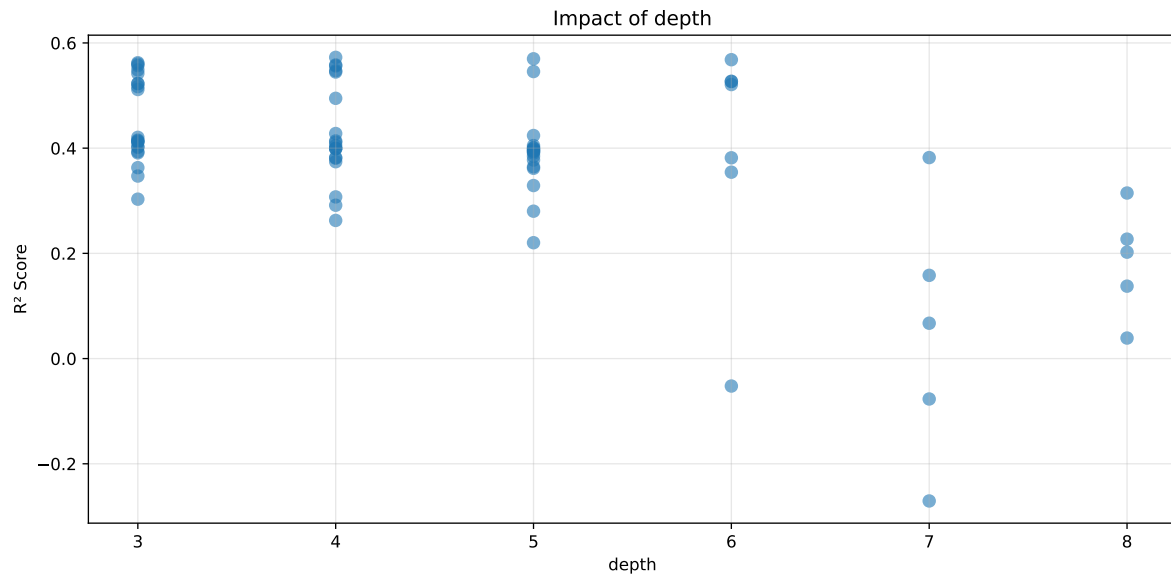
2.5.1 Gridsearch

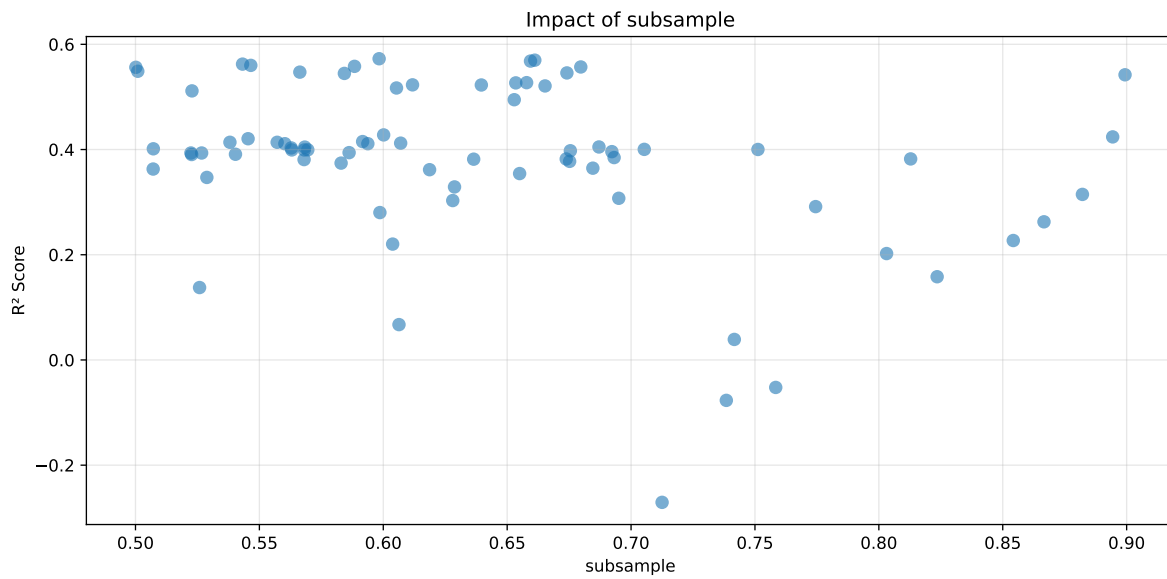
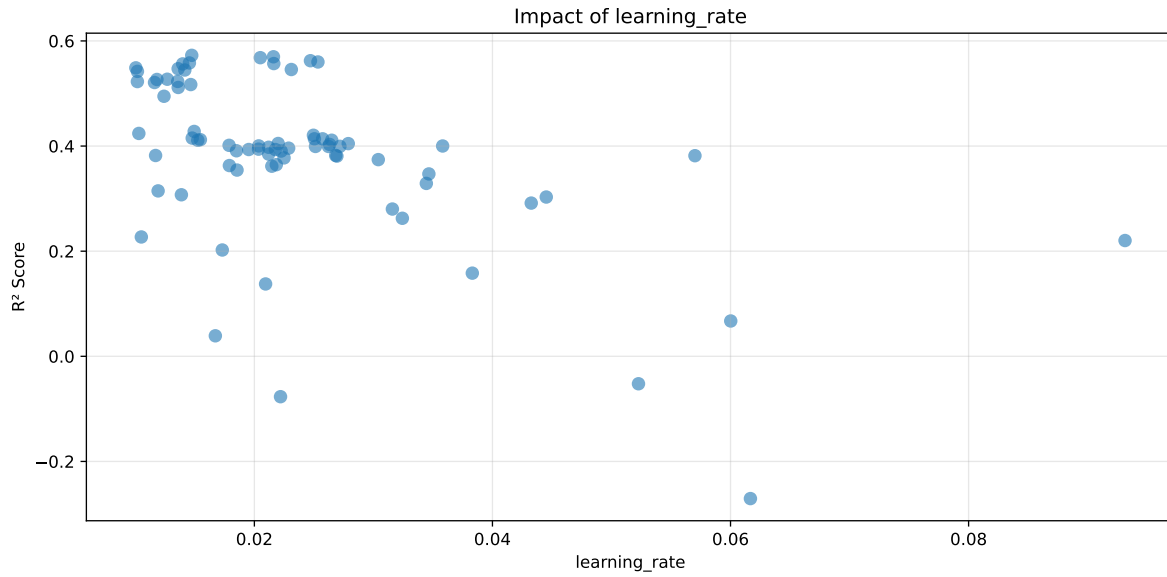
Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 43.888298 r_CO2 18.724837 r_NH3 16.359976 r_THI 11.135215
r_temperature 9.891675



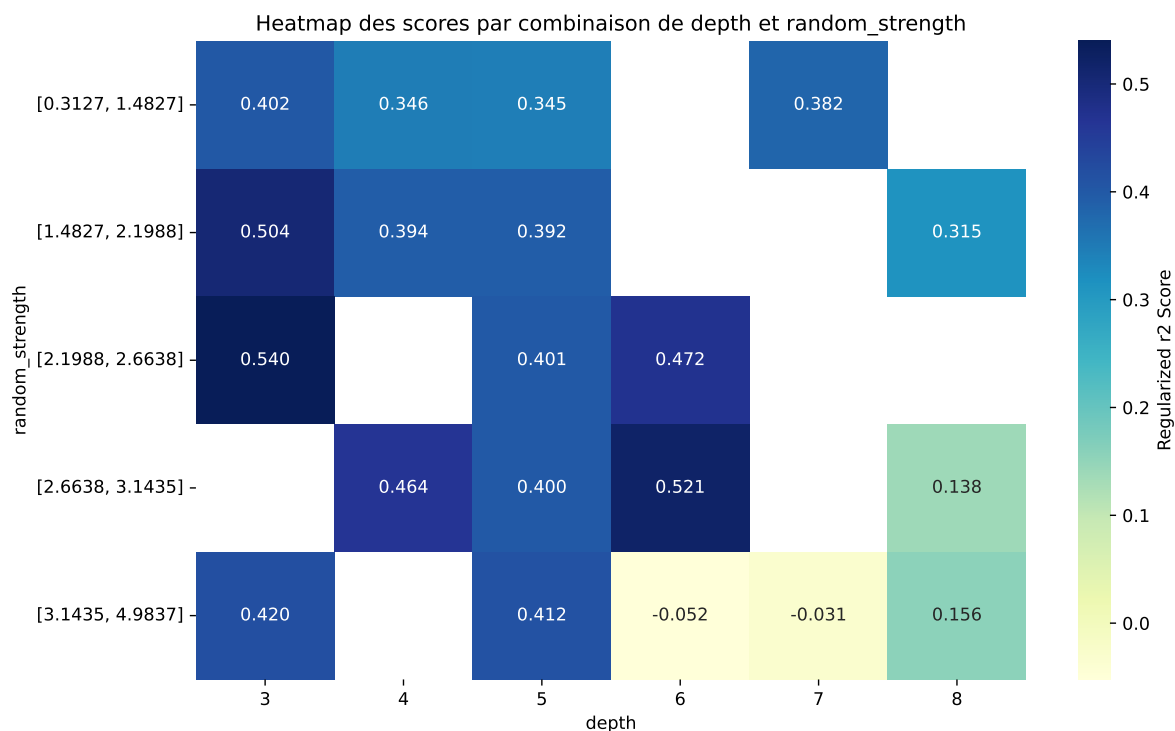
Paramètres avec >5% d'importance : ['depth', 'random_strength', 'learning_rate', 'subsample']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6011	0.5726	0.2801	0.0221	0.5726	0.2935	-0.2791	0.5126	1.0498
2	0.5966	0.5699	0.2721	0.0211	0.5699	0.2744	-0.2954	0.4816	1.0469
3	0.5927	0.5681	0.2747	0.0181	0.5681	0.2754	-0.2928	0.4847	1.0431
4	0.5875	0.5624	0.2742	0.0227	0.5624	0.2841	-0.2783	0.5051	1.0447
5	0.5844	0.56	0.2753	0.0206	0.56	0.2812	-0.2787	0.5022	1.0436
6	0.5823	0.5581	0.2694	0.0212	0.5581	0.28	-0.2781	0.5017	1.0434
7	0.5811	0.5569	0.2759	0.0204	0.5569	0.2884	-0.2685	0.5179	1.0434
8	0.5772	0.5563	0.2727	0.0203	0.5563	0.2837	-0.2726	0.51	1.0375
9	0.5684	0.5488	0.2826	0.0186	0.5488	0.2847	-0.2641	0.5188	1.0357
10	0.565	0.5471	0.2836	0.018	0.5471	0.2827	-0.2645	0.5166	1.0327
---	---	---	---	---	---	---	---	---	---
75	0.8437	0.658	0.2164	0.0213	-0.2707	0.2215	-0.4365	0.3366	1.2823

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.014739310560307283, 'reg__l2_leaf_reg': 44.46879549407582, 'reg__bagging_temperature': 1.297628598749451, 'reg__random_strength': 2.696246383342957, 'reg__subsample': 0.5983237927433019, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7598794570917619, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2121.244037457623}

Rank 2: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.021602517873083964, 'reg__l2_leaf_reg': 28.6852249702257, 'reg__bagging_temperature': 1.4576129825756112, 'reg__random_strength': 2.1745404442556615, 'reg__subsample': 0.6611513684451449, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7503431980989415, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 933.4040699750936}

Rank 3: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.0205000337106323, 'reg__l2_leaf_reg': 41.12613212062385, 'reg__bagging_temperature': 0.8592527601690405, 'reg__random_strength': 2.454863034257345, 'reg__subsample': 0.6594195633552139, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7970810695395925, 'reg__border_count': 80, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3171.4742042000707}

Rank 4: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.02471177118052754, 'reg__l2_leaf_reg': 38.76678498213383, 'reg__bagging_temperature': 2.106171609288598, 'reg__random_strength': 1.6613103062767958, 'reg__subsample': 0.5432112252264382, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7287072200910424, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 108.64374776070144}

Rank 5: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.025341145312731985, 'reg__l2_leaf_reg': 39.67007932387948, 'reg__bagging_temperature': 2.0094120641449607, 'reg__random_strength': 1.7613211673785998, 'reg__subsample': 0.546536334169004, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7251341628905608, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 56.376541317360385}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01454183493434941, 'reg__l2_leaf_reg': 33.96971774864267, 'reg__bagging_temperature': 1.1117657938180041, 'reg__random_strength': 2.827892393924721, 'reg__subsample': 0.588390045032016, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7674866291301838, 'reg__border_count': 44, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1858.825920855474}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.021631471388385056, 'reg__l2_leaf_reg': 39.793942371794216, 'reg__bagging_temperature': 1.7551287078131121, 'reg__random_strength': 2.6562560632288976, 'reg__subsample': 0.6797077824127401,

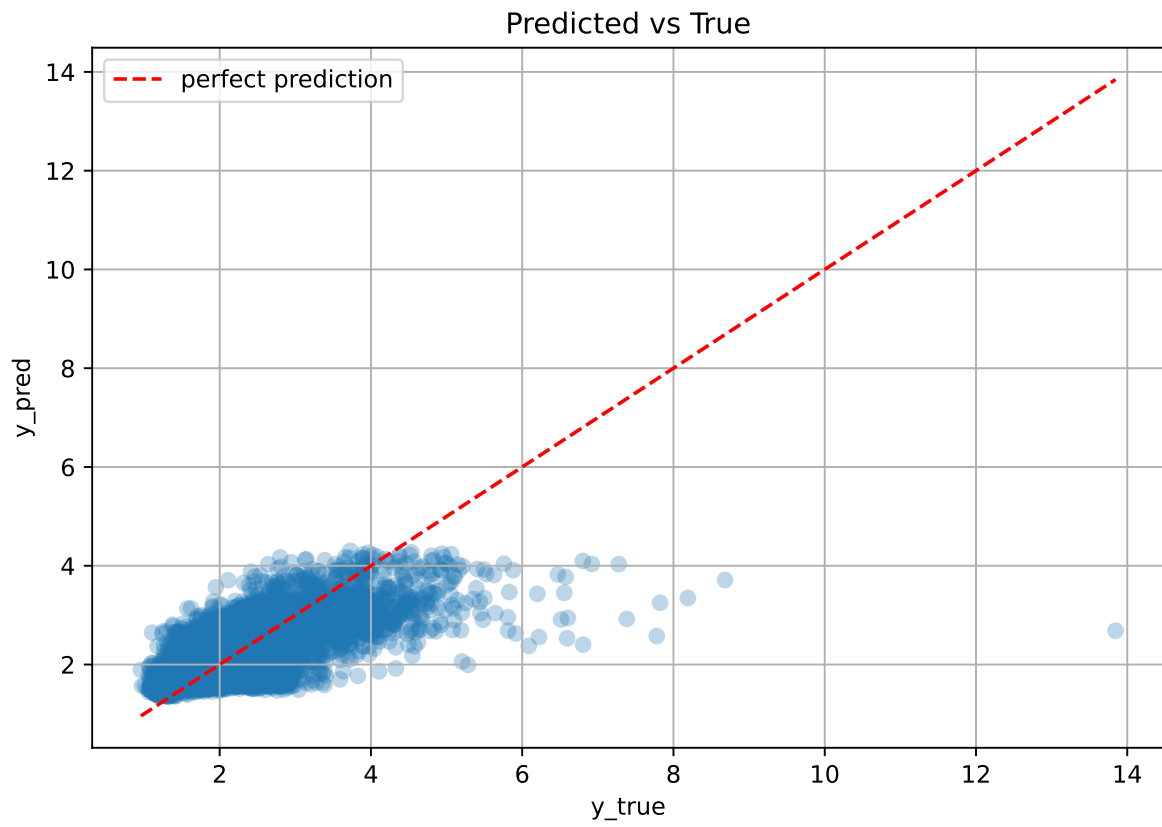
'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6977306556860227, 'reg__border_count': 75, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2799.5433245788927}

Rank 8: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.013978333018615707, 'reg__l2_leaf_reg': 33.98199585870937, 'reg__bagging_temperature': 1.1787265601310248, 'reg__random_strength': 2.8394137691818466, 'reg__subsample': 0.5001409246998193, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7589318545190076, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1907.016530759886}

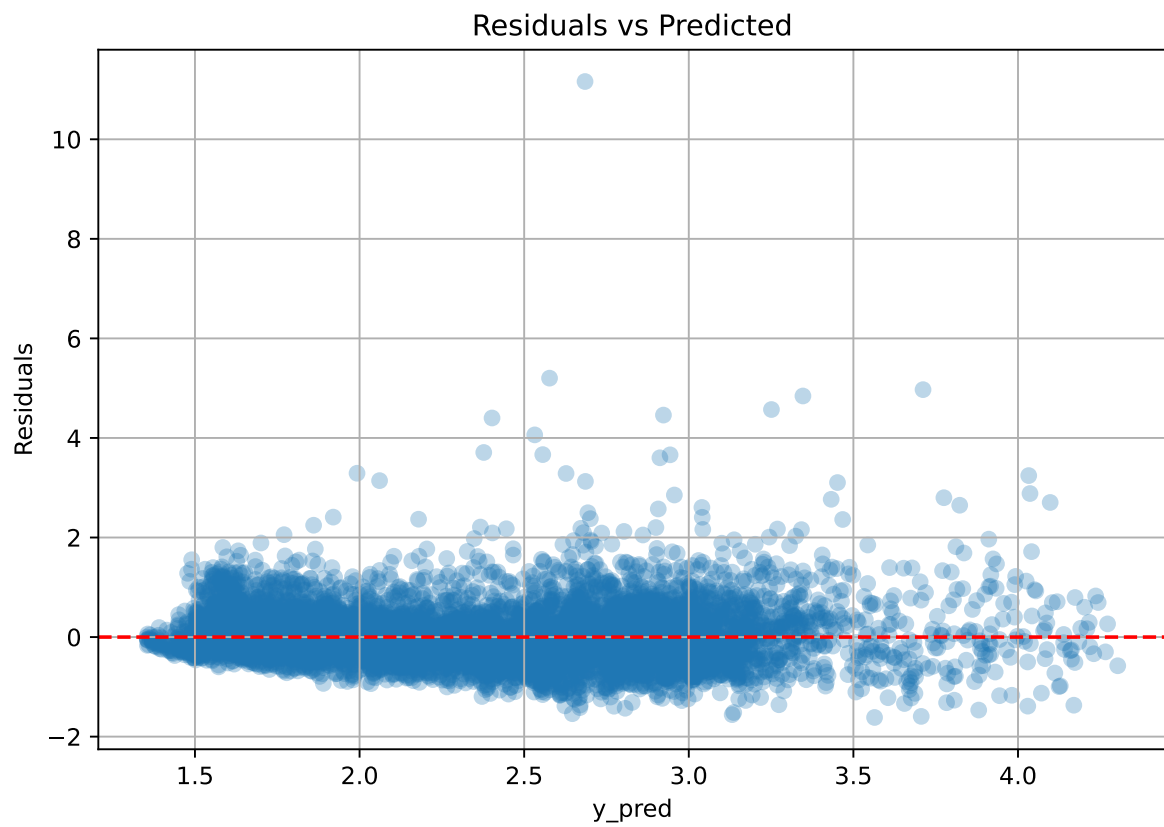
Rank 9: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.01004248082153229, 'reg__l2_leaf_reg': 47.22731115619695, 'reg__bagging_temperature': 2.73918245364022, 'reg__random_strength': 3.5842729679433436, 'reg__subsample': 0.5009235095915306, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.6334635666226973, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9957.143316586022}

Rank 10: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.01359709151775628, 'reg__l2_leaf_reg': 45.05770664867538, 'reg__bagging_temperature': 2.232885654434793, 'reg__random_strength': 2.9088210764150233, 'reg__subsample': 0.5663223592347952, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6808776660106818, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2009.572807759309}

2.5.2 Scatter



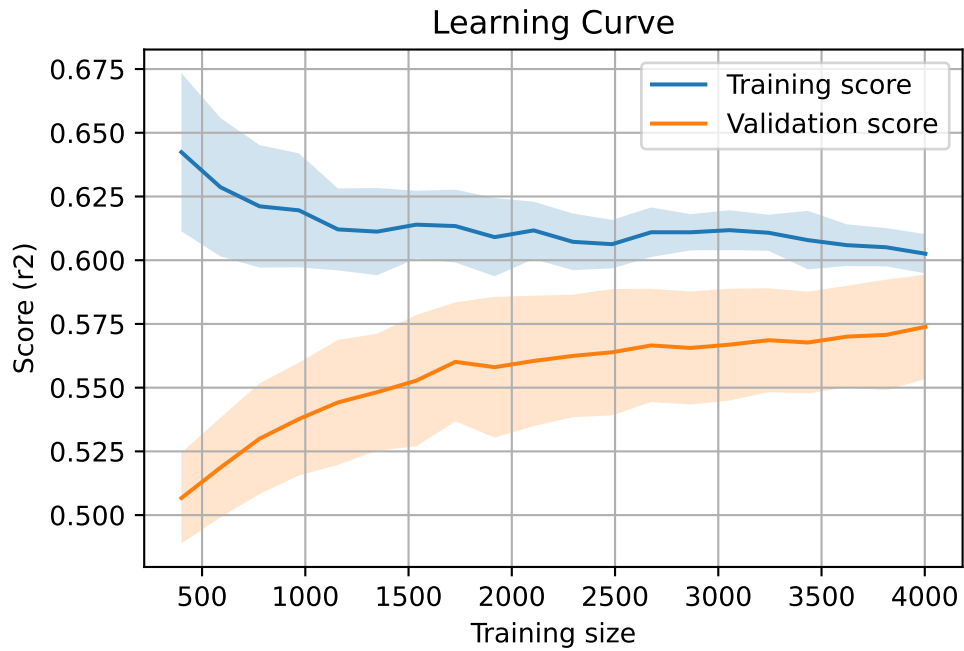
2.5.3 Scatter residuals



2.5.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.403	0.408	0.011	0.187	0.187	-0.222	0.457	0.987	nan
LGBM	0.578	0.55	0.019	0.274	0.266	-0.284	0.484	1.05	0.5505
CatBoost	0.601	0.573	0.022	0.28	0.294	-0.279	0.513	1.05	0.5726

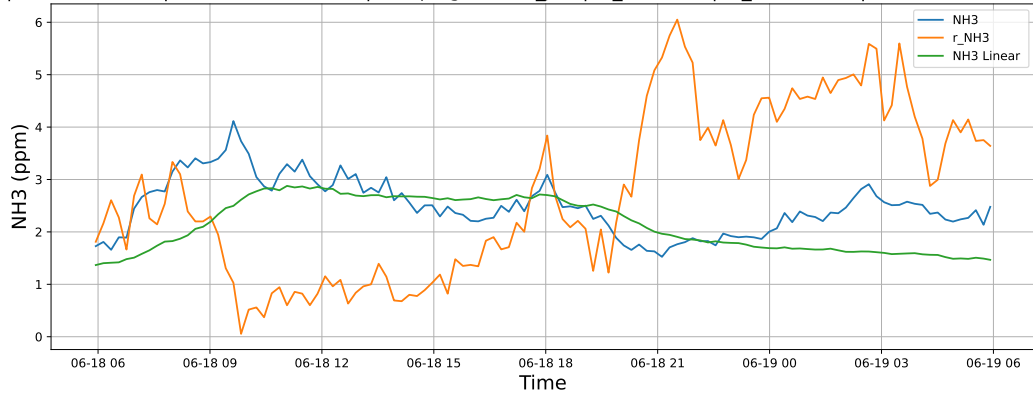
3 Plot

3.1 Standalone plots

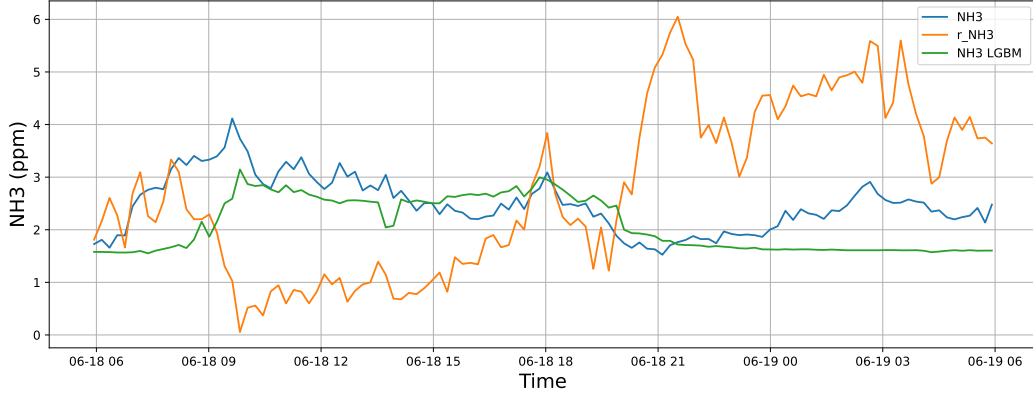
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

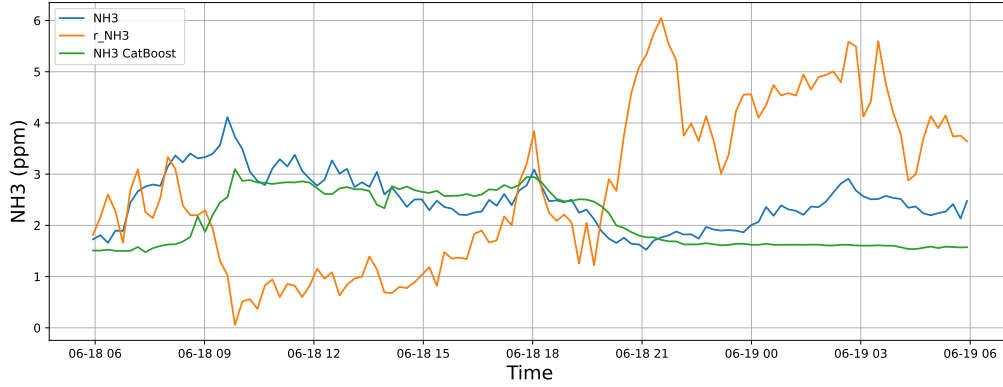
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

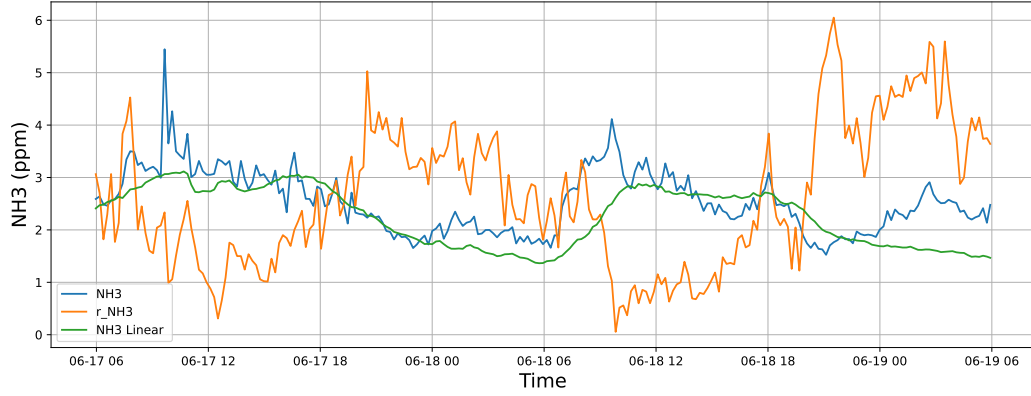


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

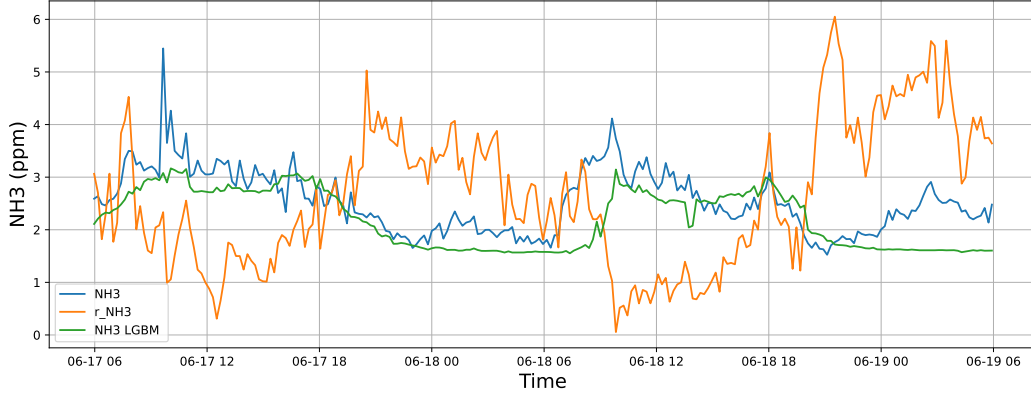


3.1.1.2 Time window : 48

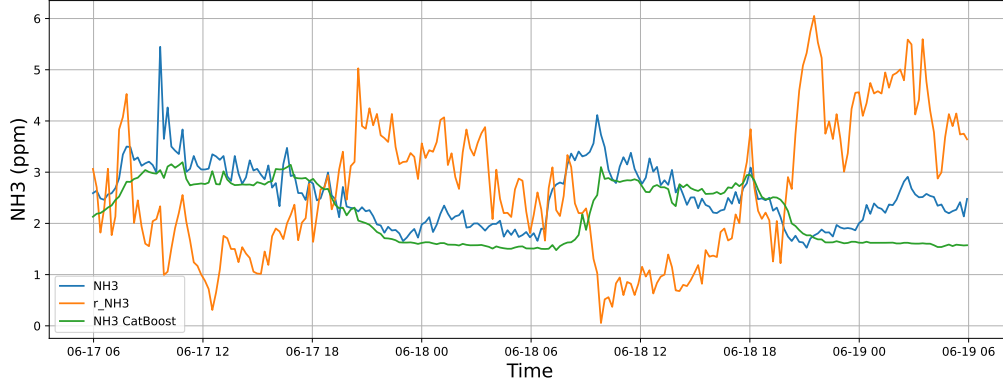
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

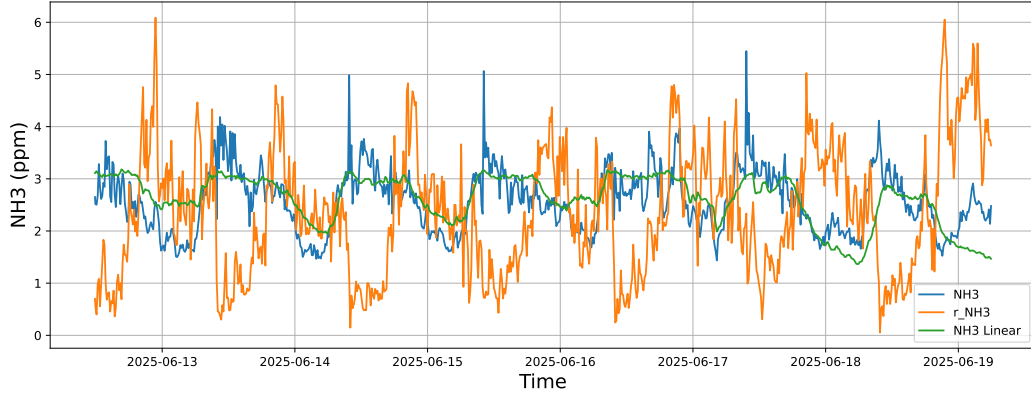


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

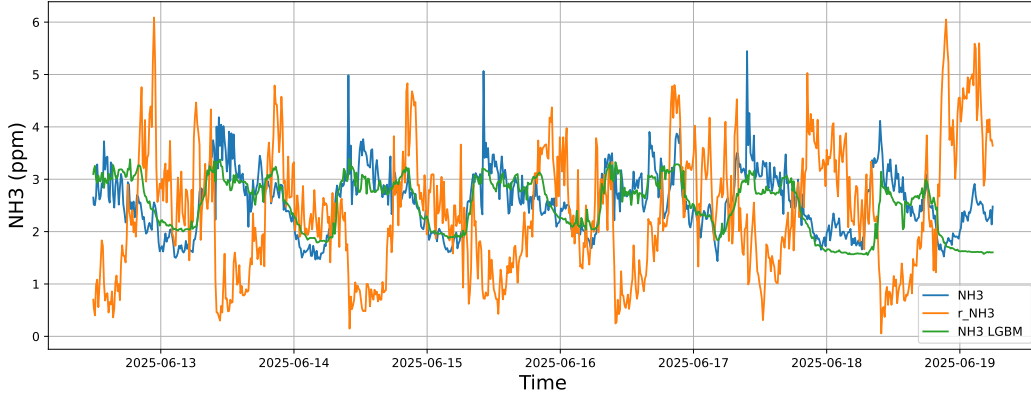


3.1.1.3 Time window : ALL

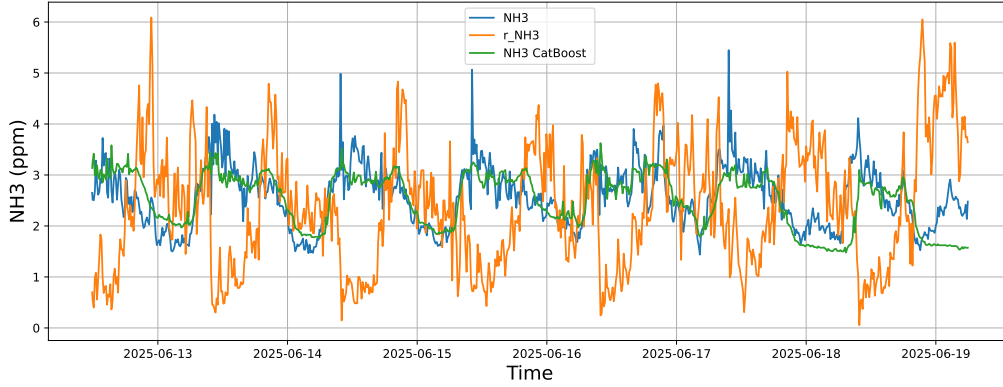
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



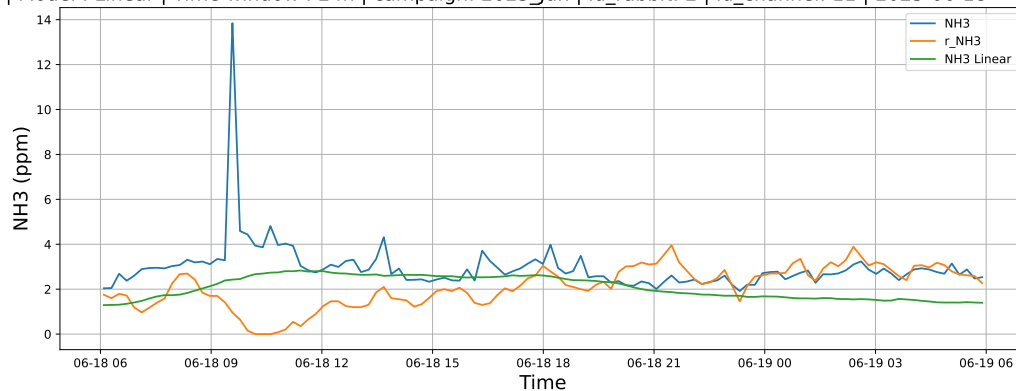
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



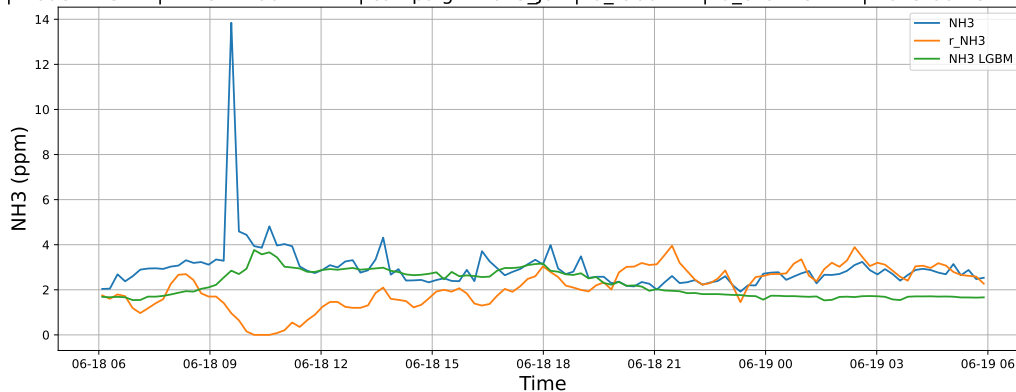
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

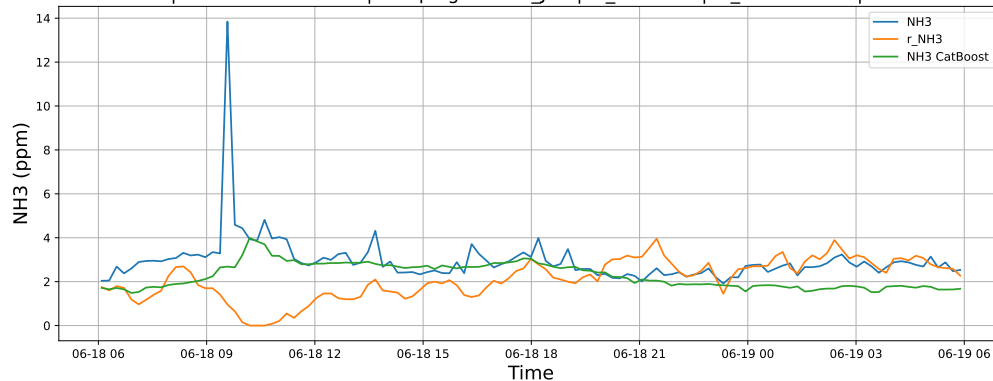
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

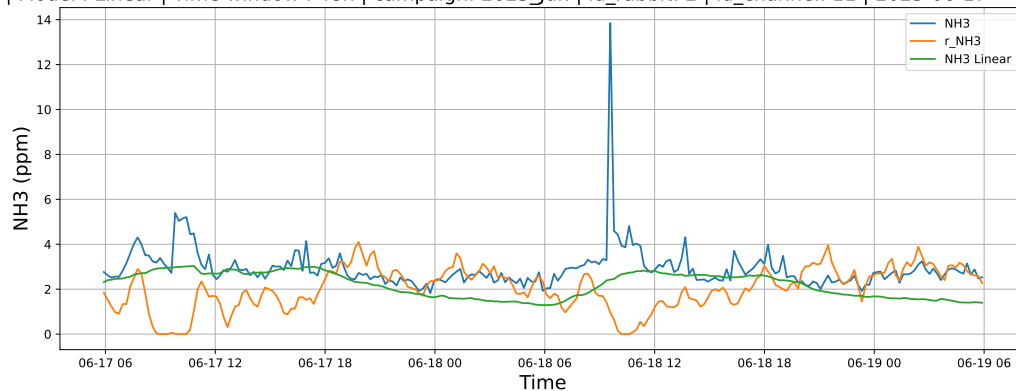


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

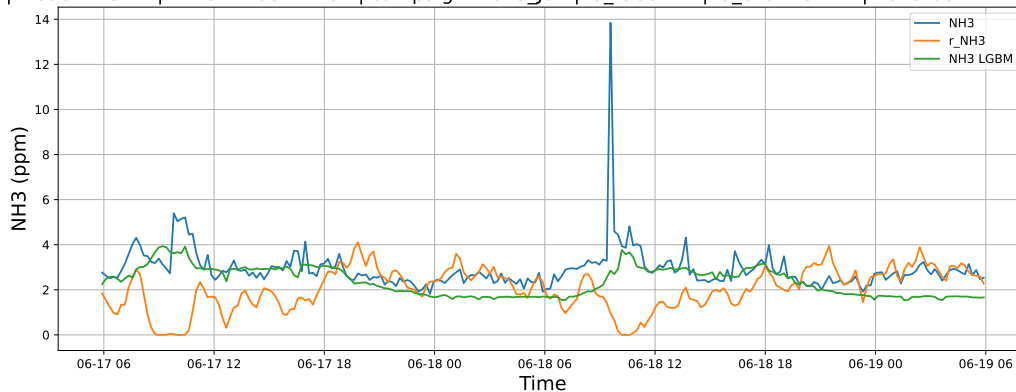


3.1.2.2 Time window : 48

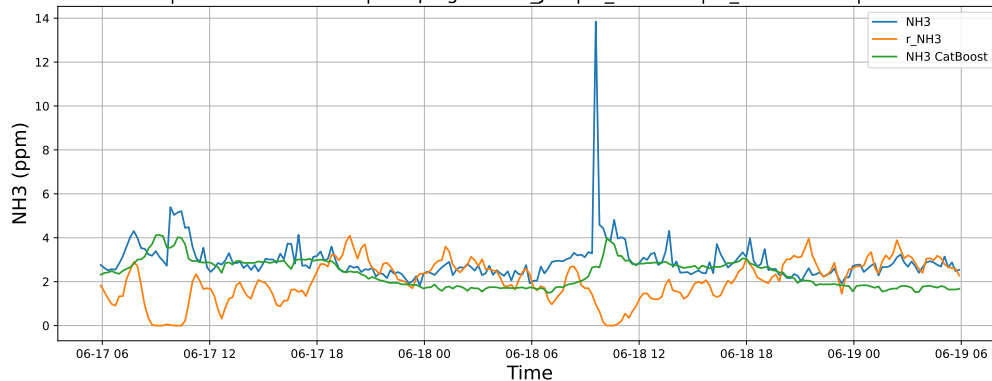
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

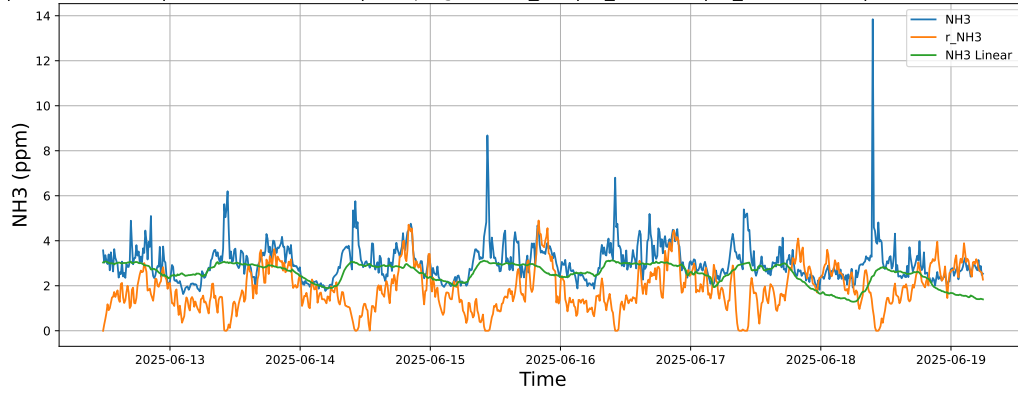


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

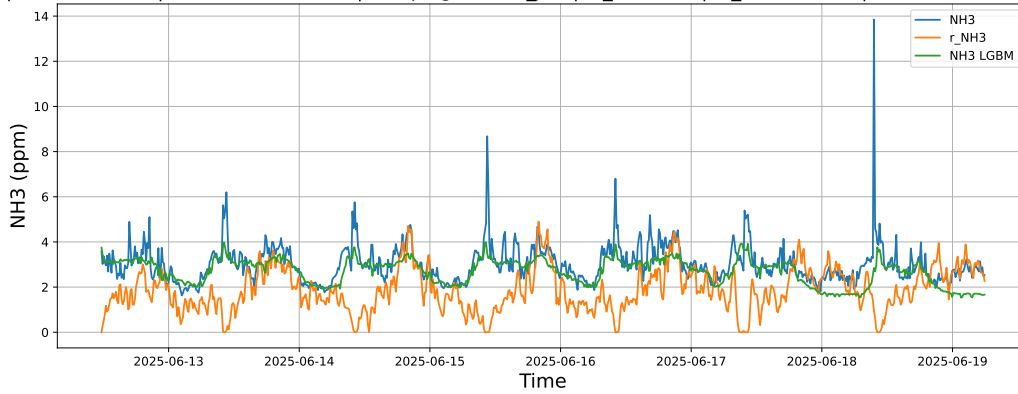


3.1.2.3 Time window : ALL

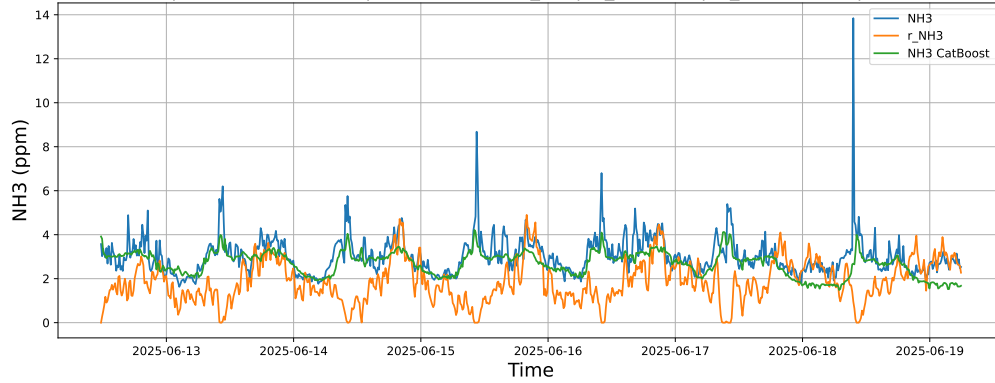
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



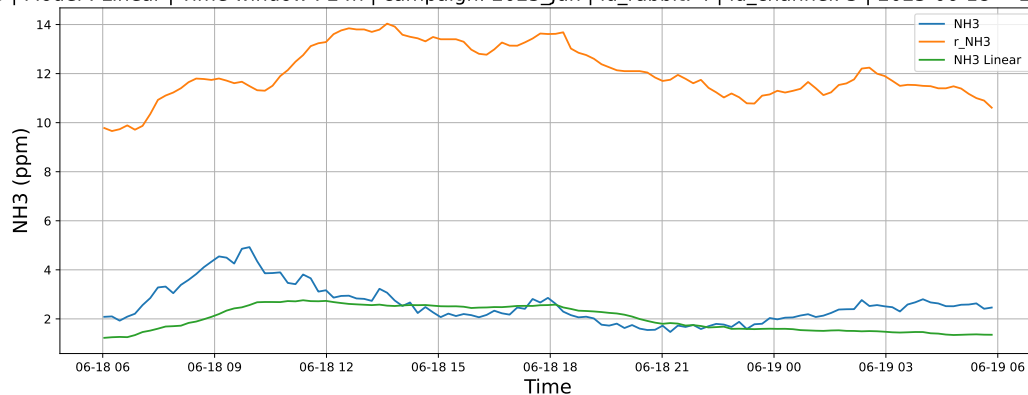
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



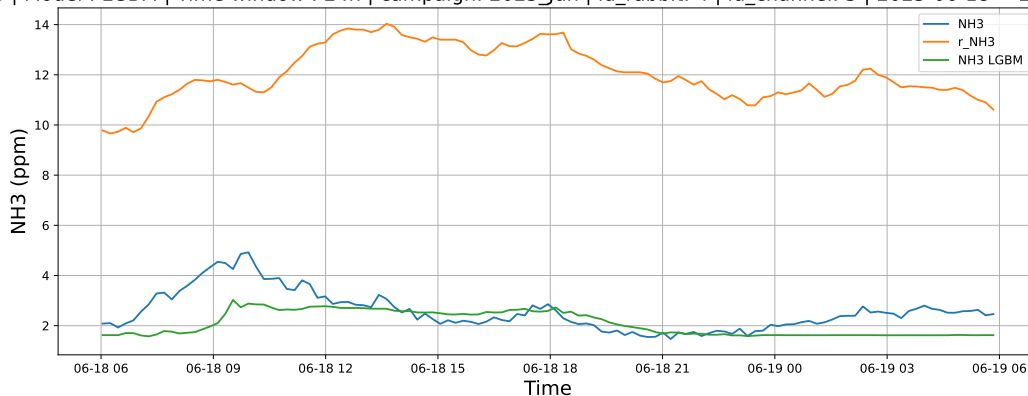
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

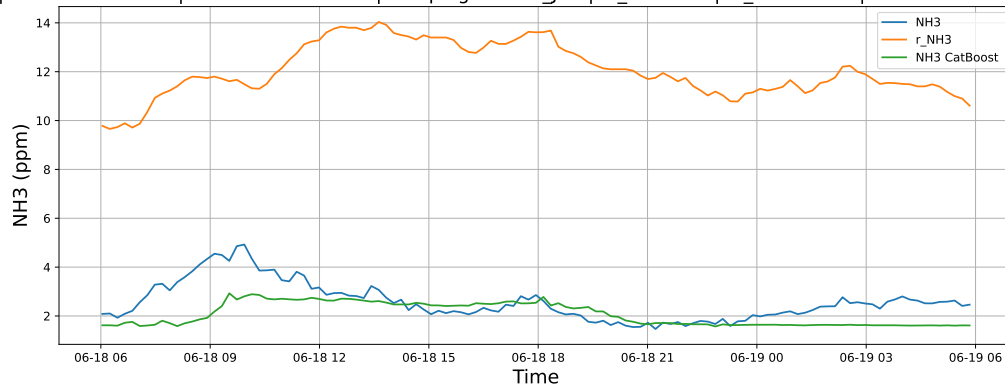
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

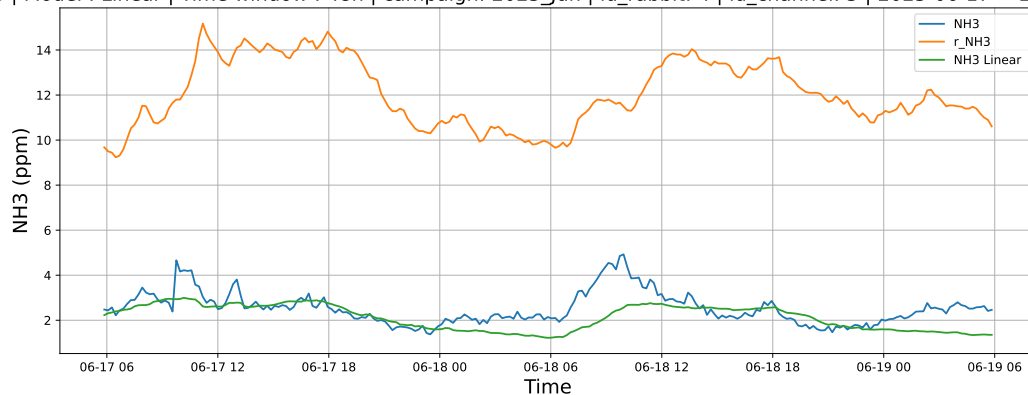


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

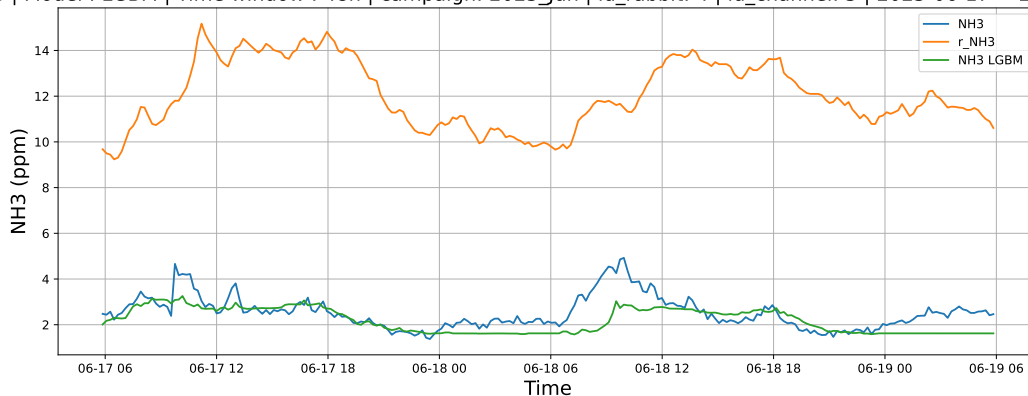


3.1.3.2 Time window : 48

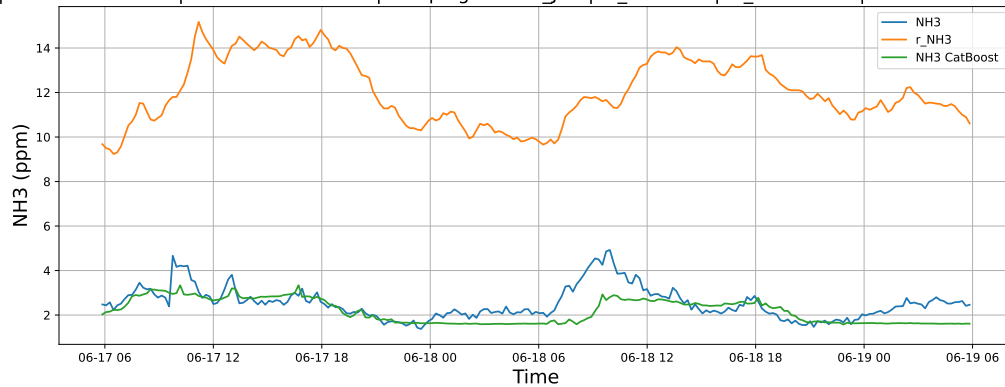
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

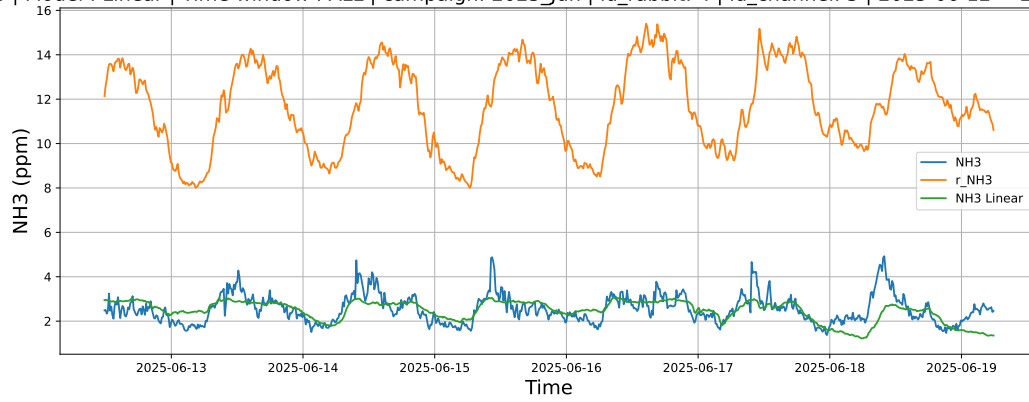


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

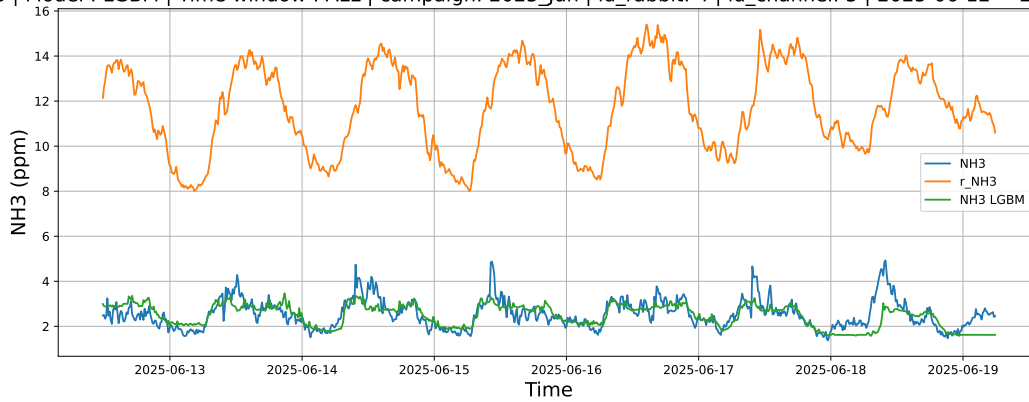


3.1.3.3 Time window : ALL

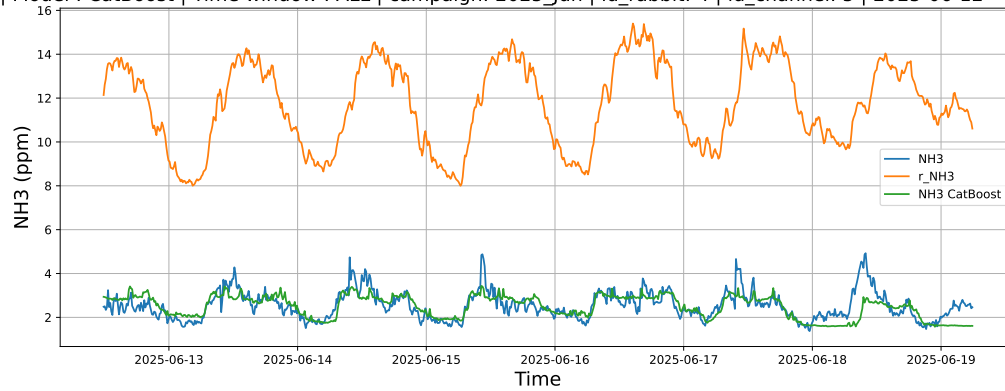
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



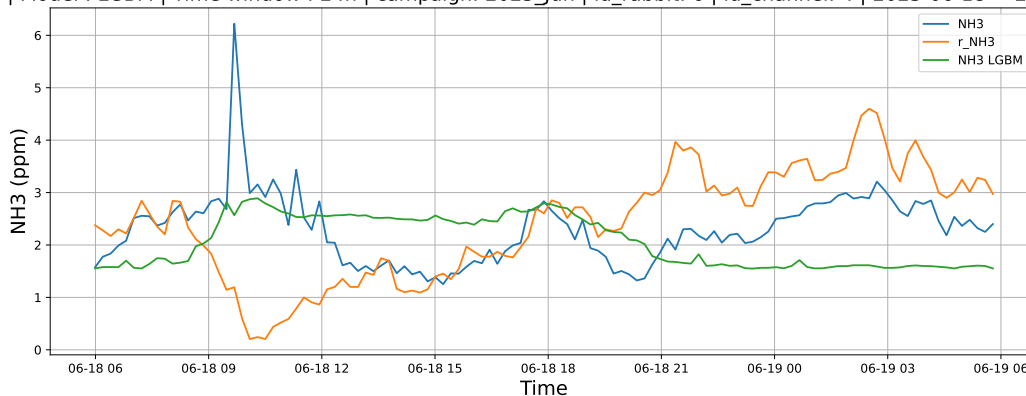
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

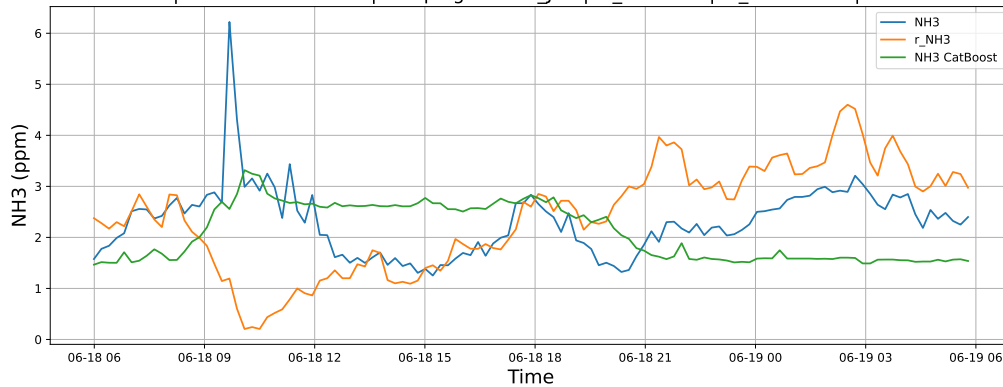
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

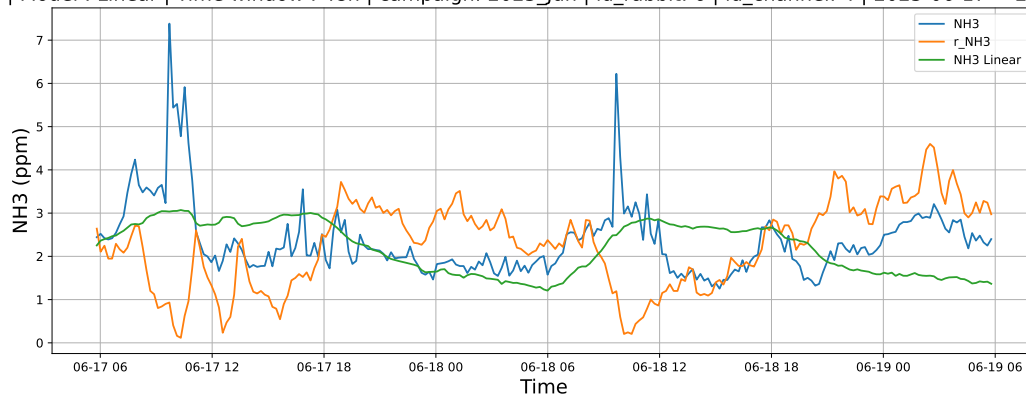


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

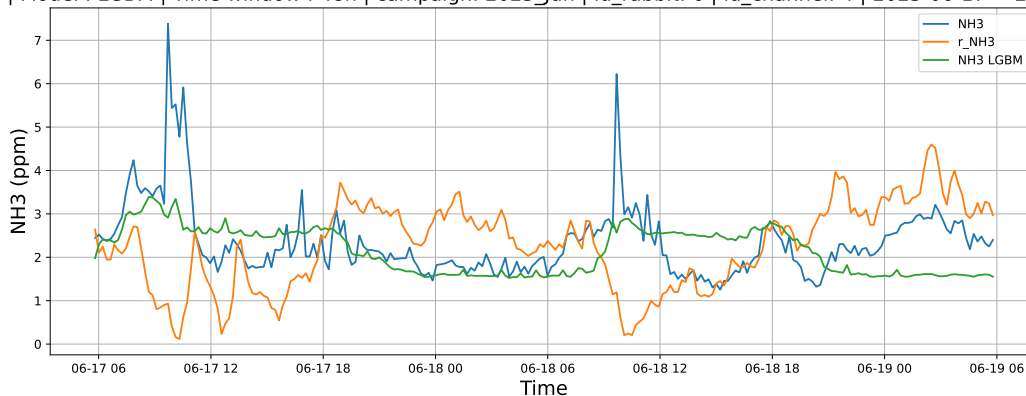


3.1.4.2 Time window : 48

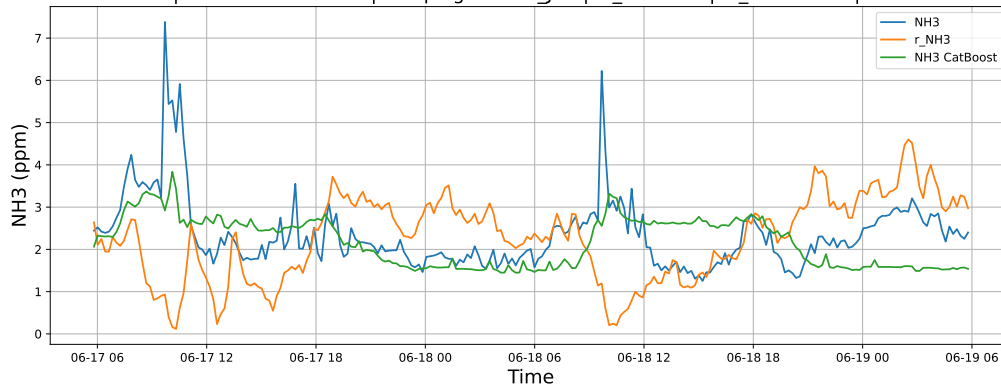
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

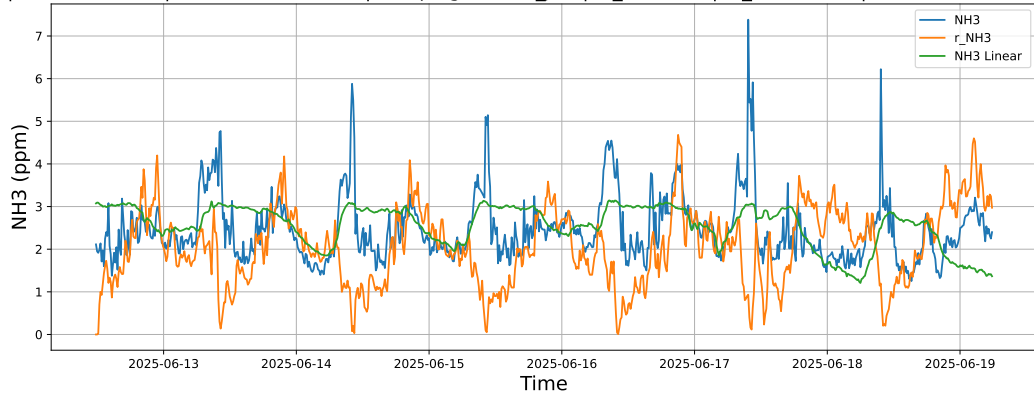


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

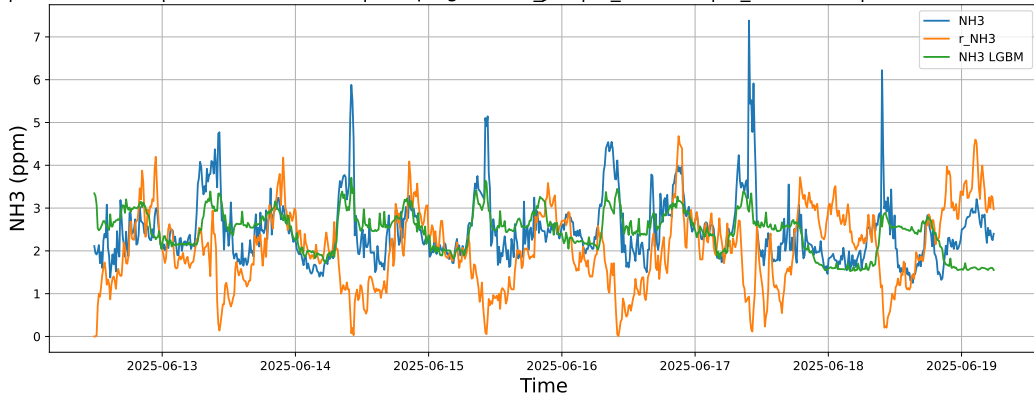


3.1.4.3 Time window : ALL

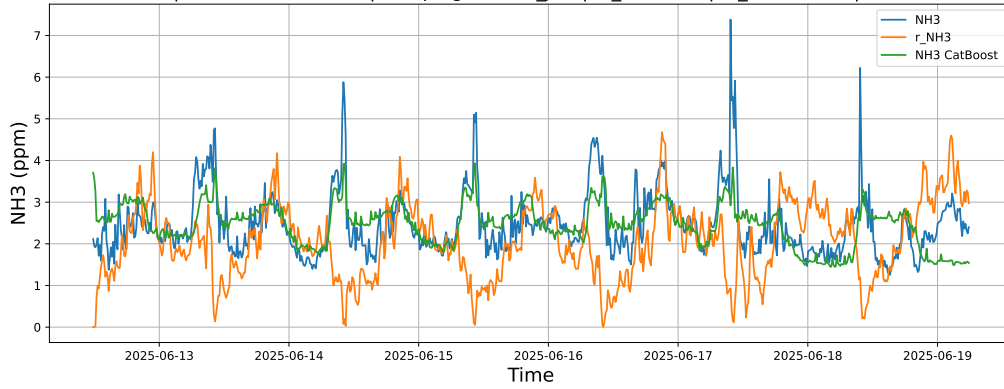
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



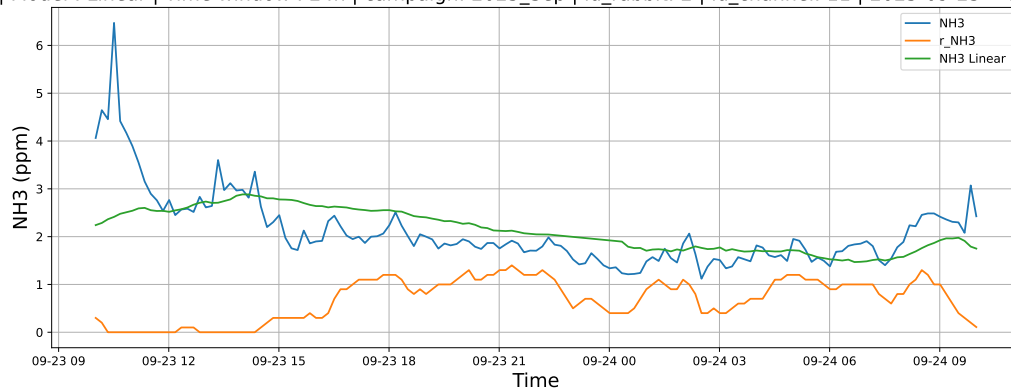
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



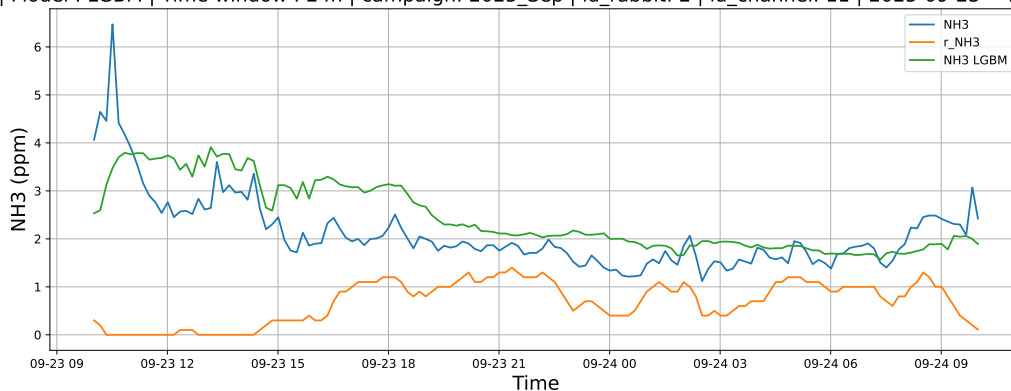
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

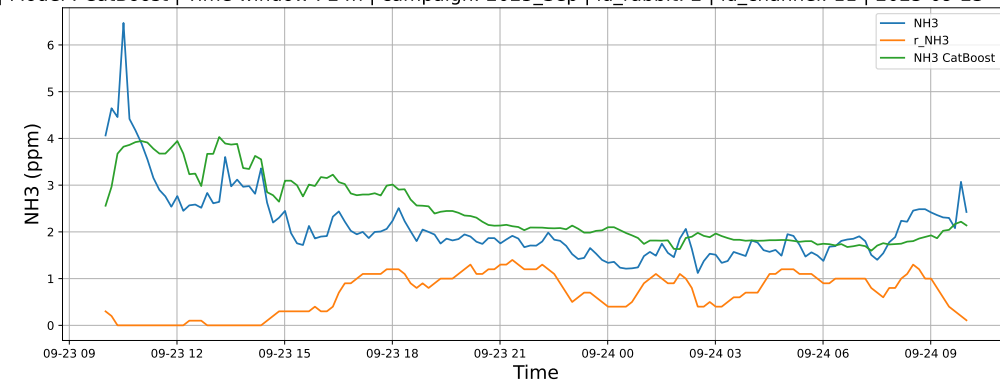
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

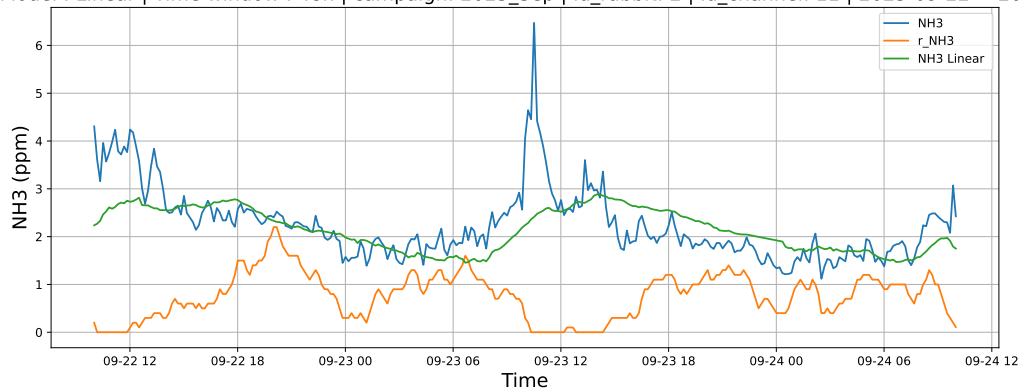


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

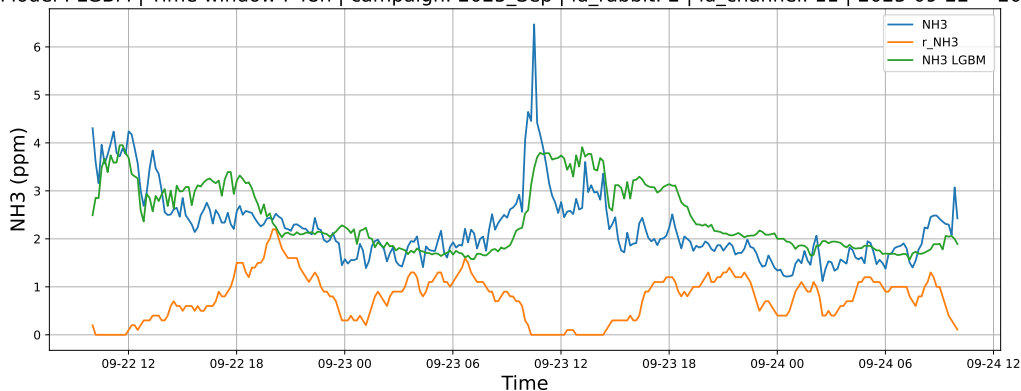


3.1.5.2 Time window : 48

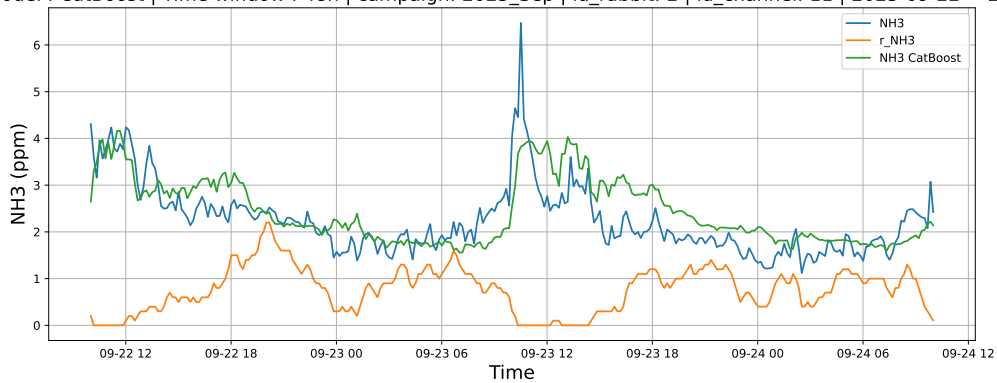
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

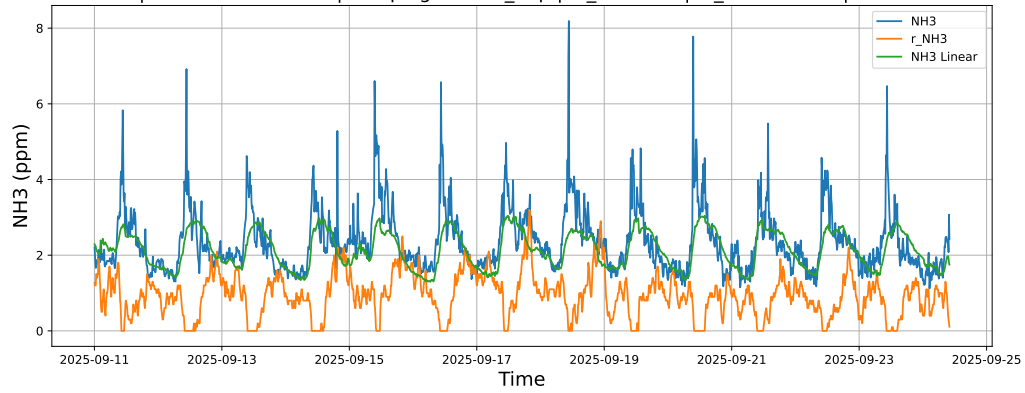


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

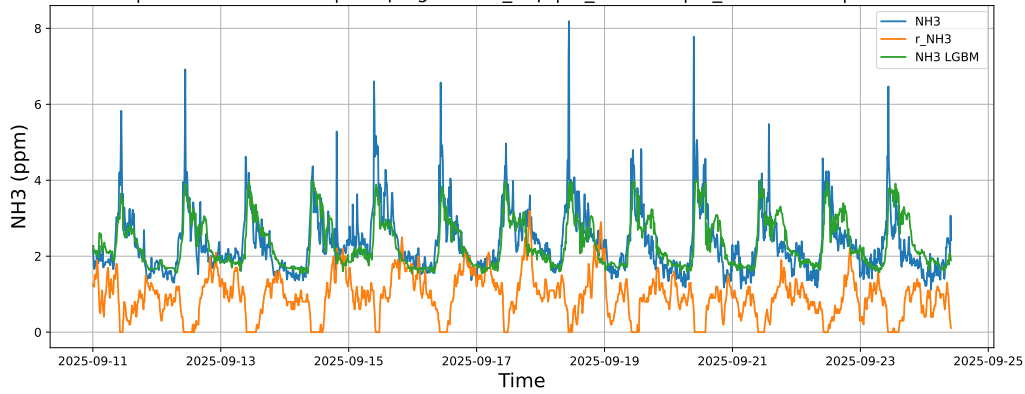


3.1.5.3 Time window : ALL

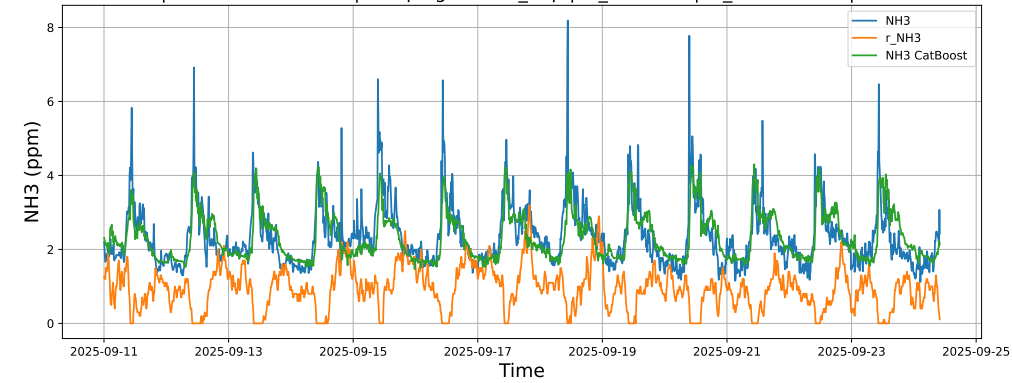
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



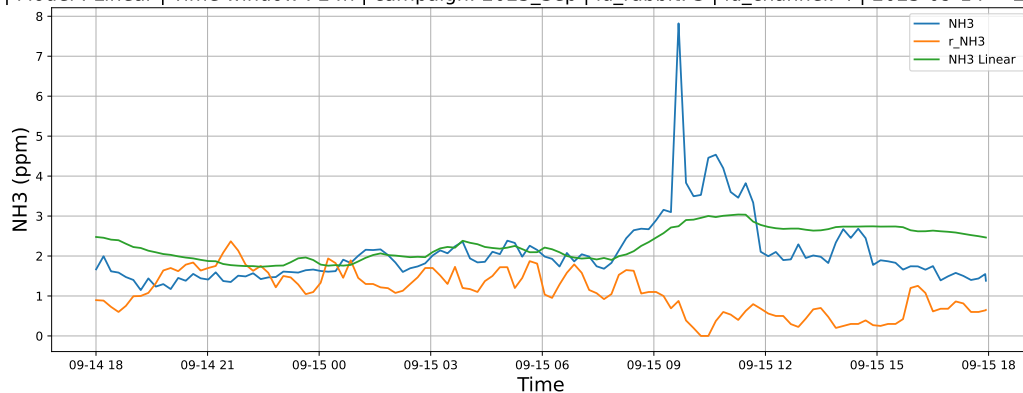
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



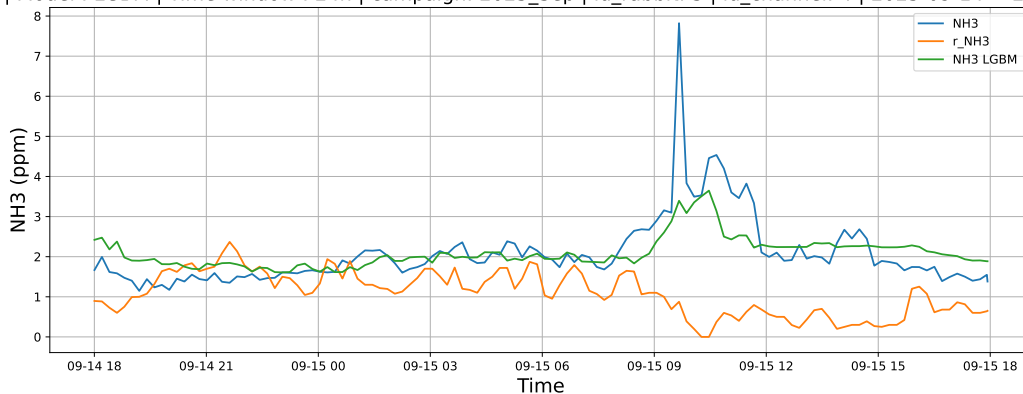
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

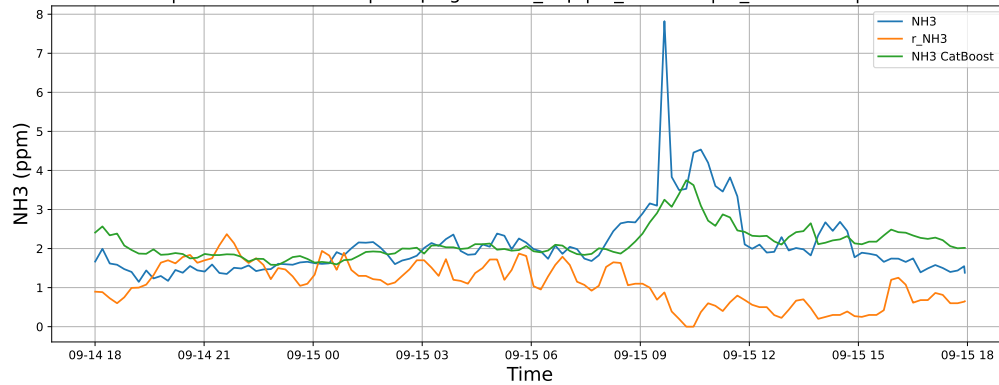
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

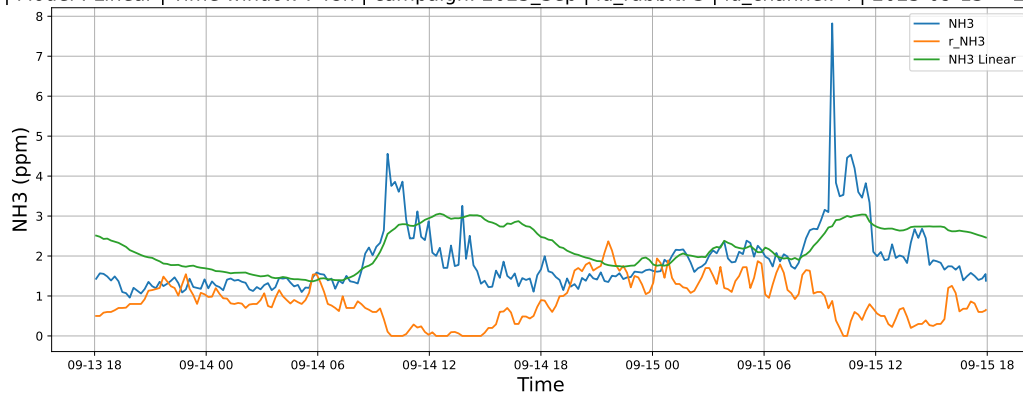


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

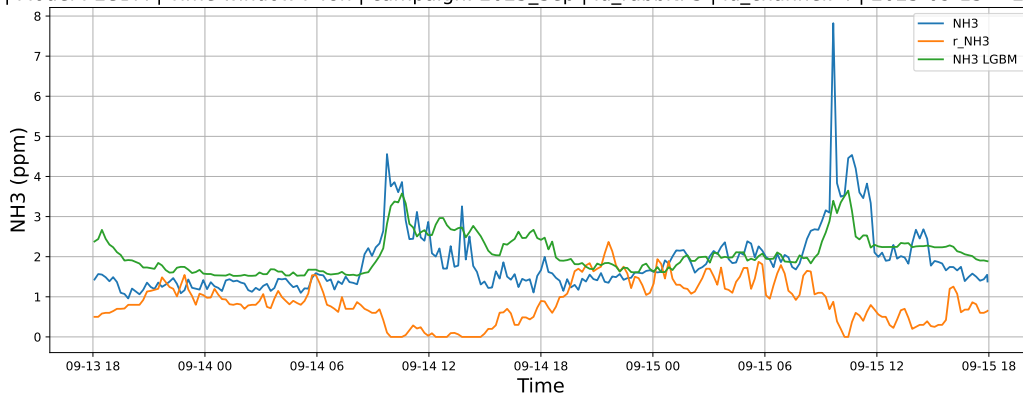


3.1.6.2 Time window : 48

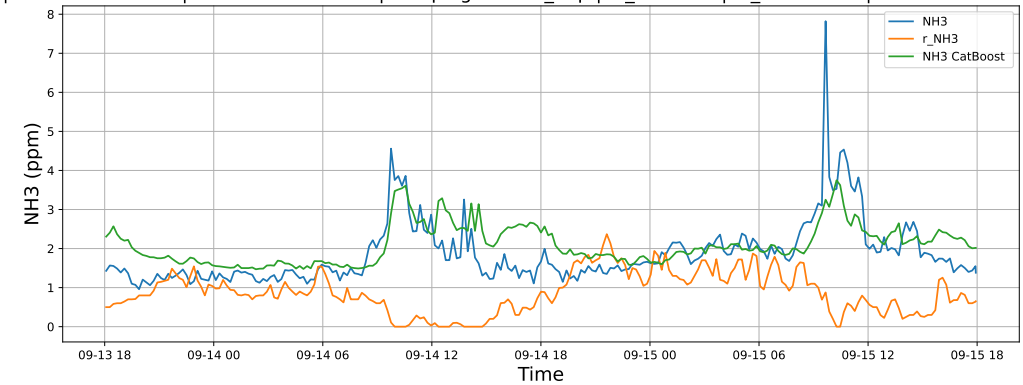
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

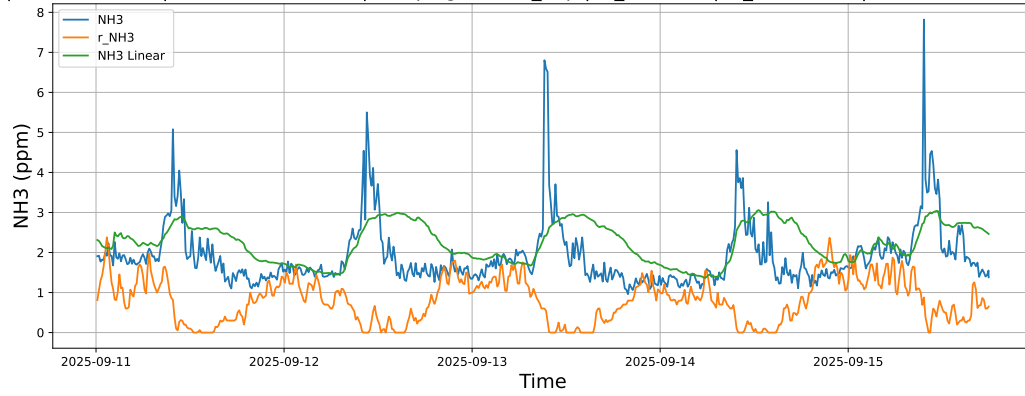


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

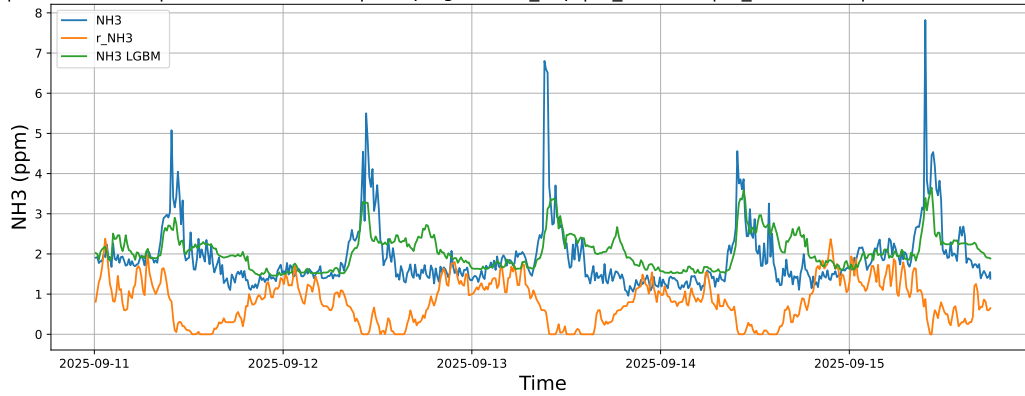


3.1.6.3 Time window : ALL

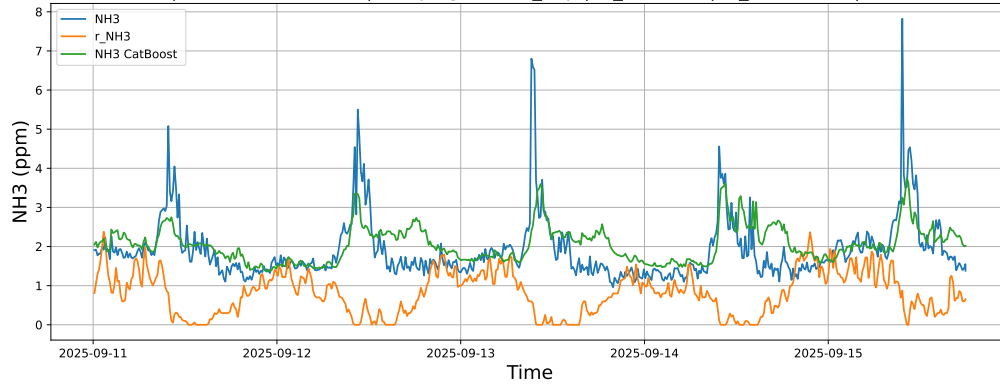
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



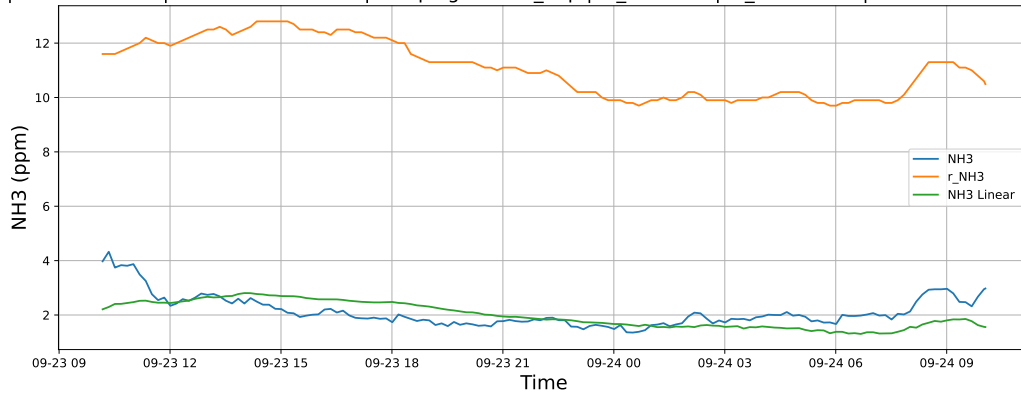
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



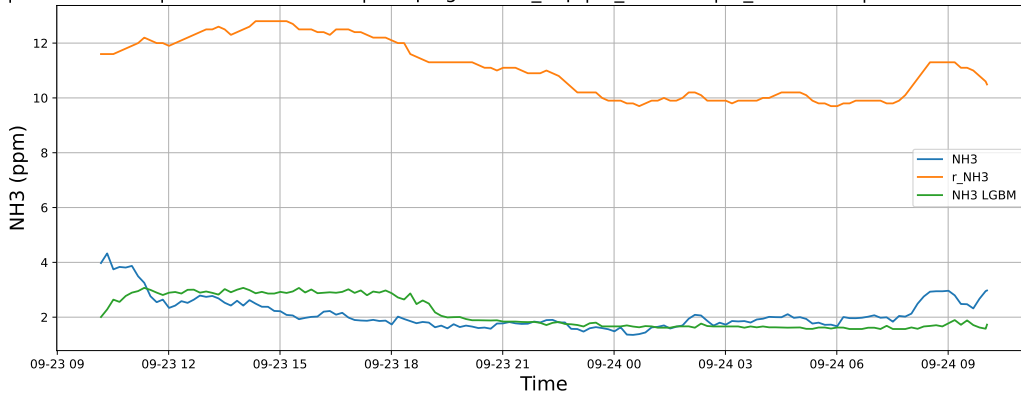
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

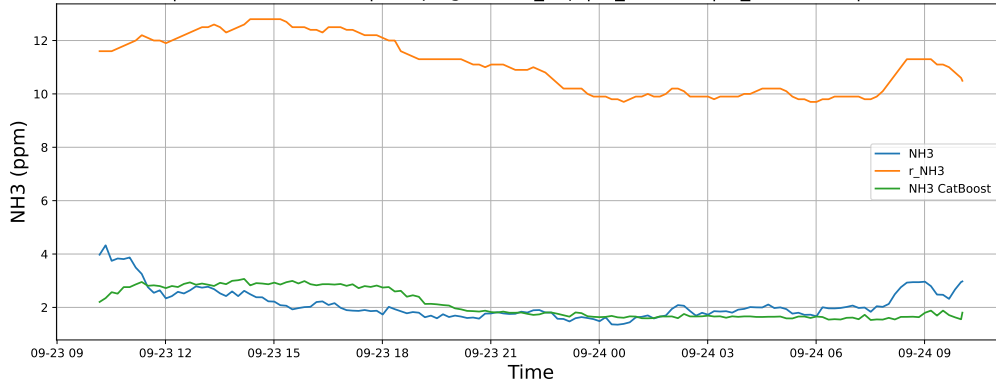
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

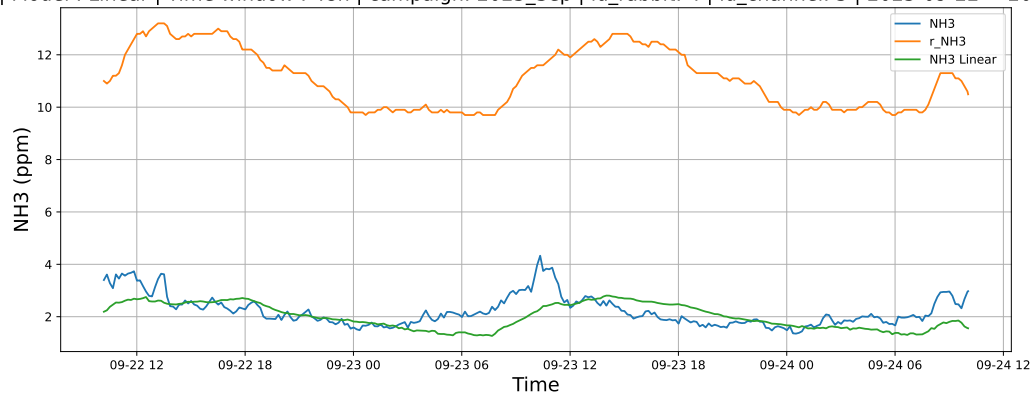


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

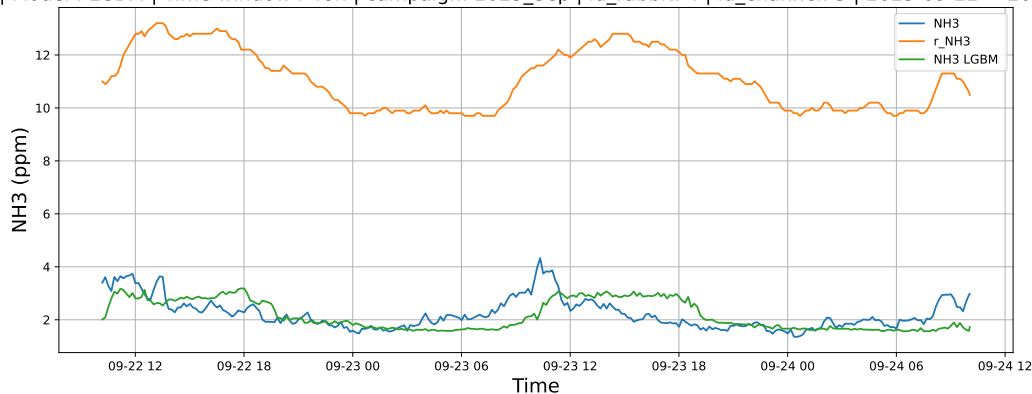


3.1.7.2 Time window : 48

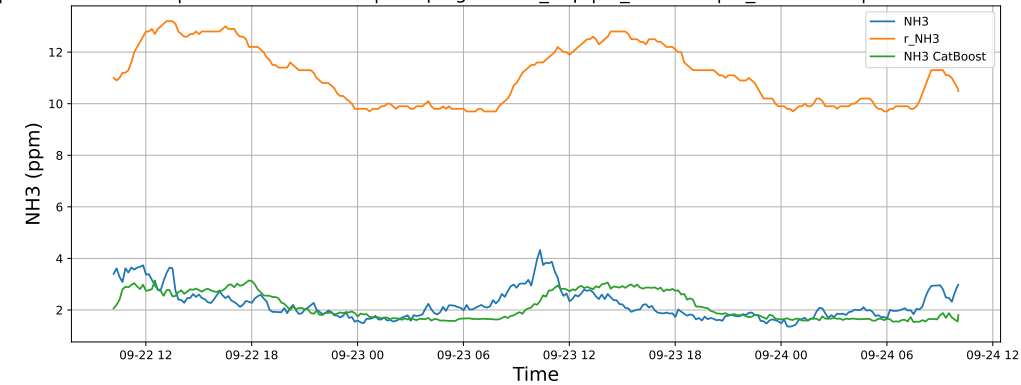
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

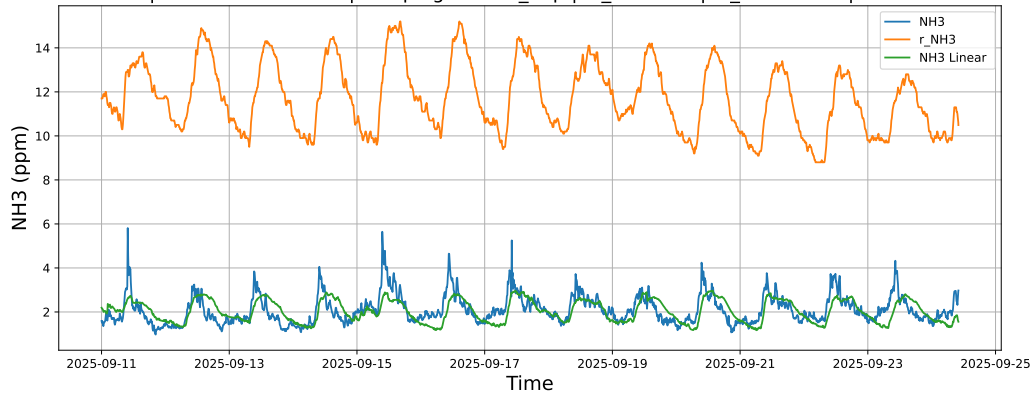


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

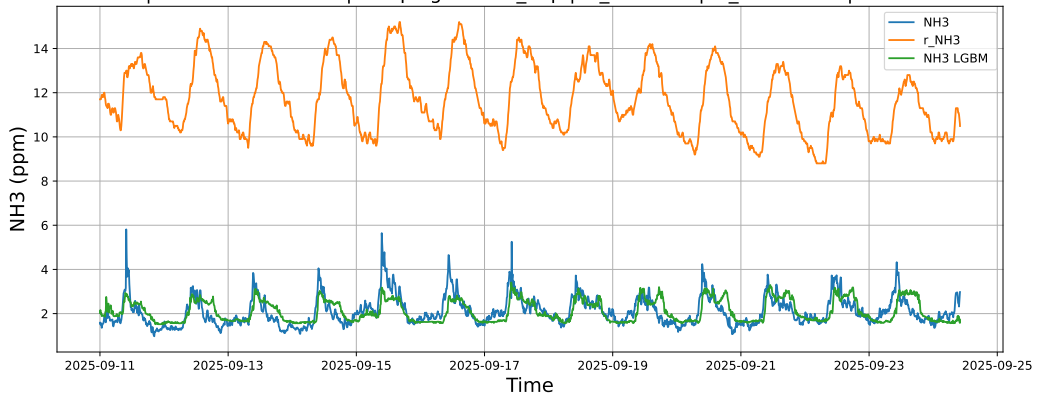


3.1.7.3 Time window : ALL

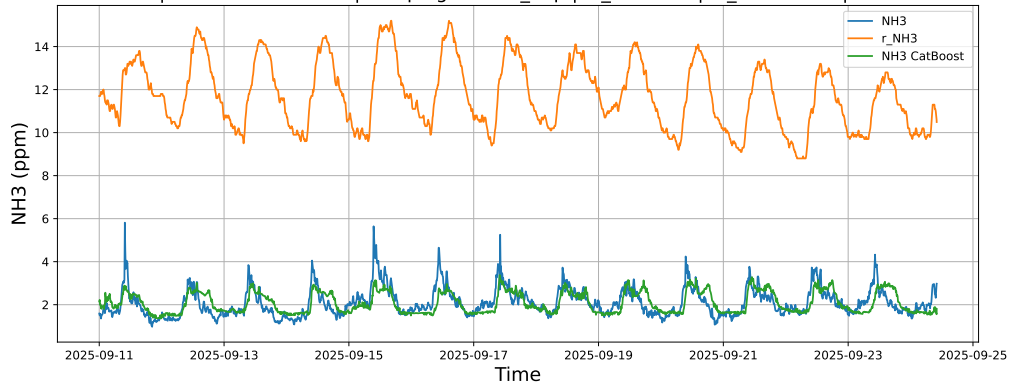
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



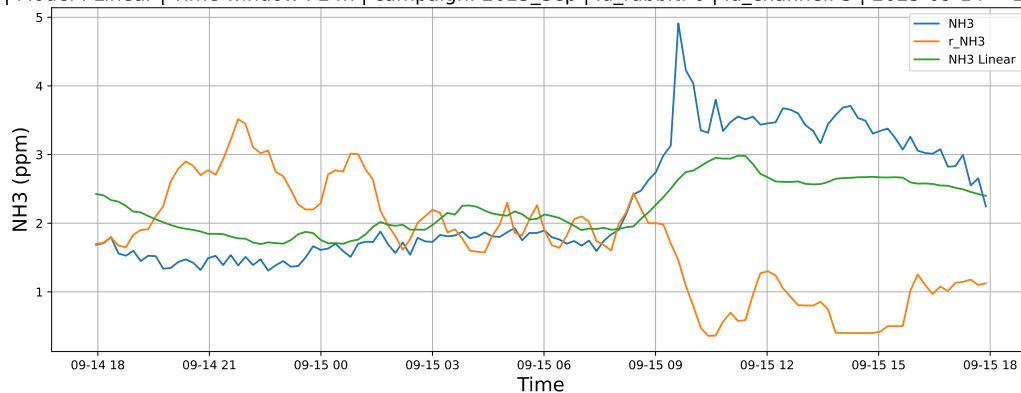
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



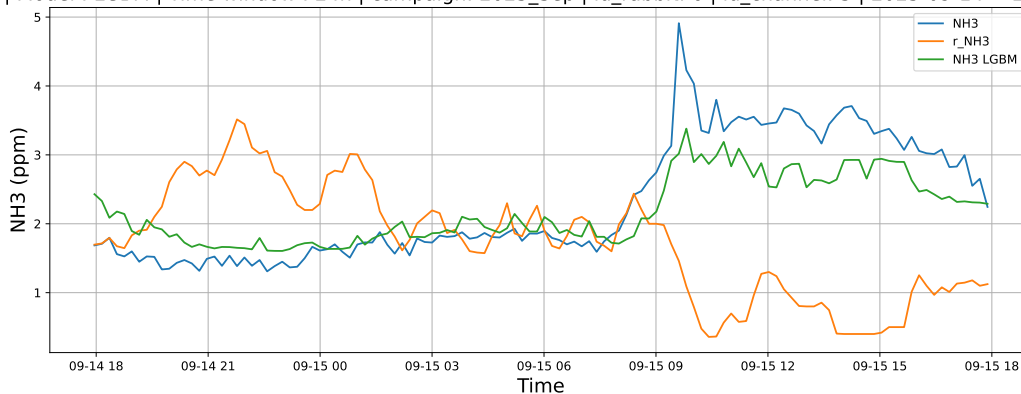
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

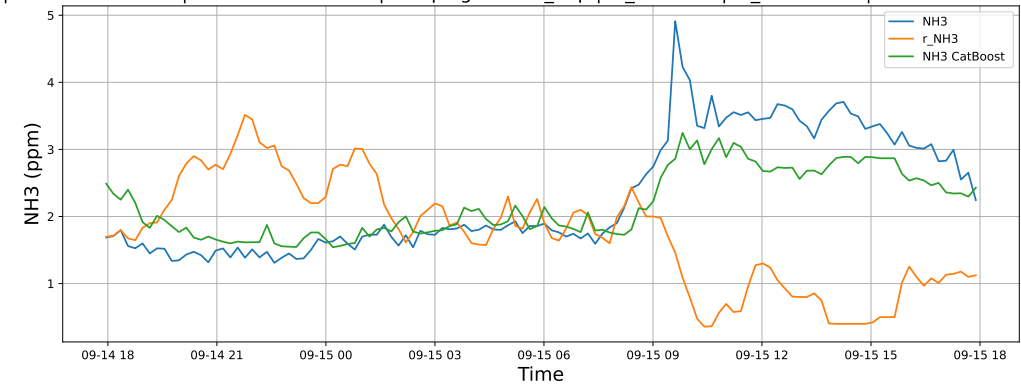
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

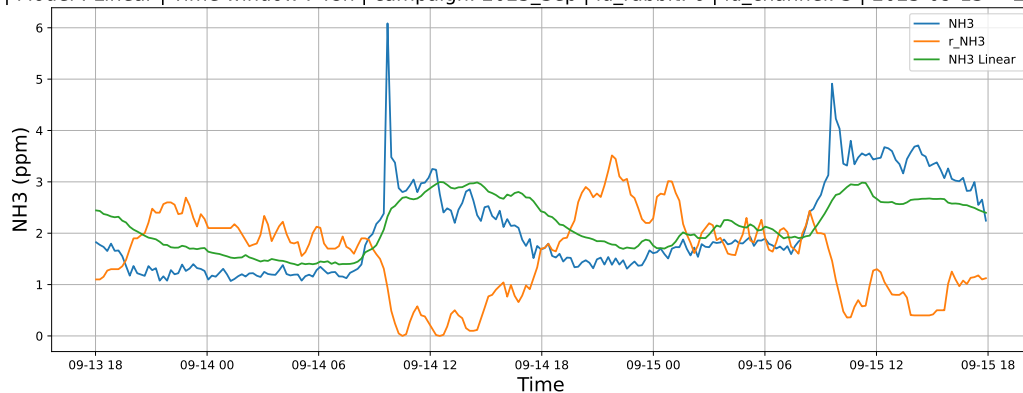


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

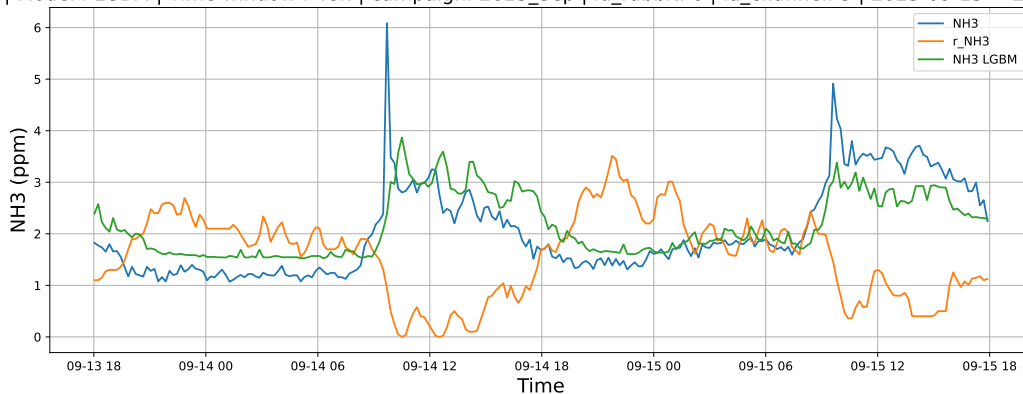


3.1.8.2 Time window : 48

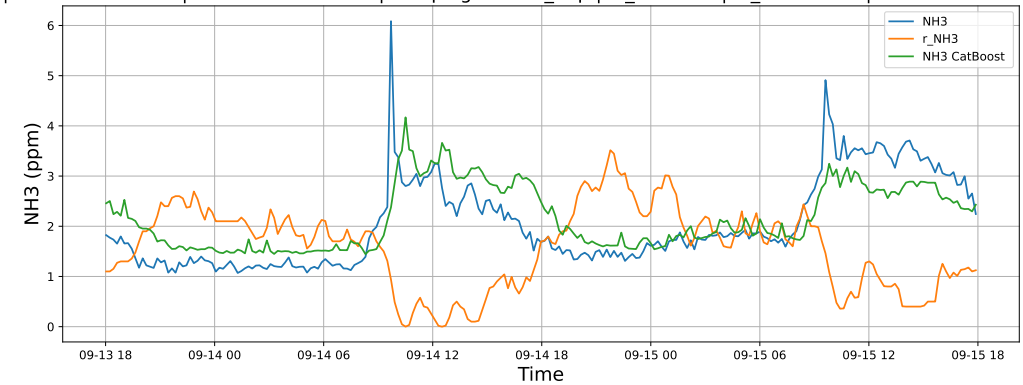
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

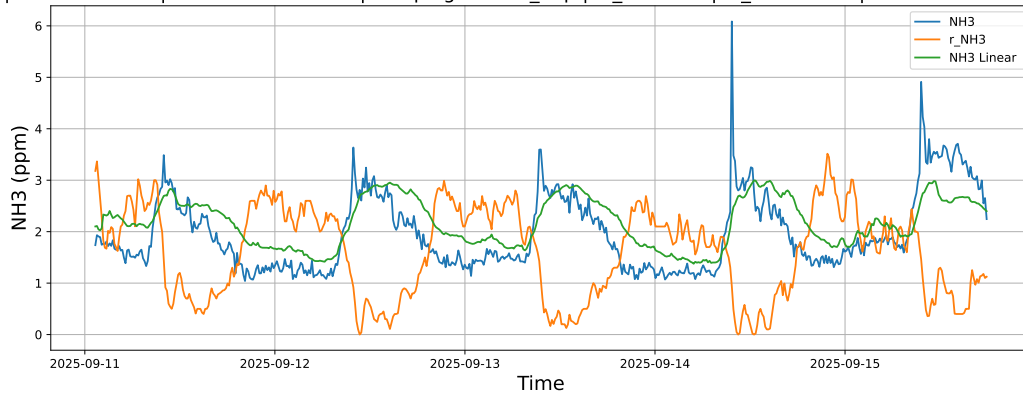


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

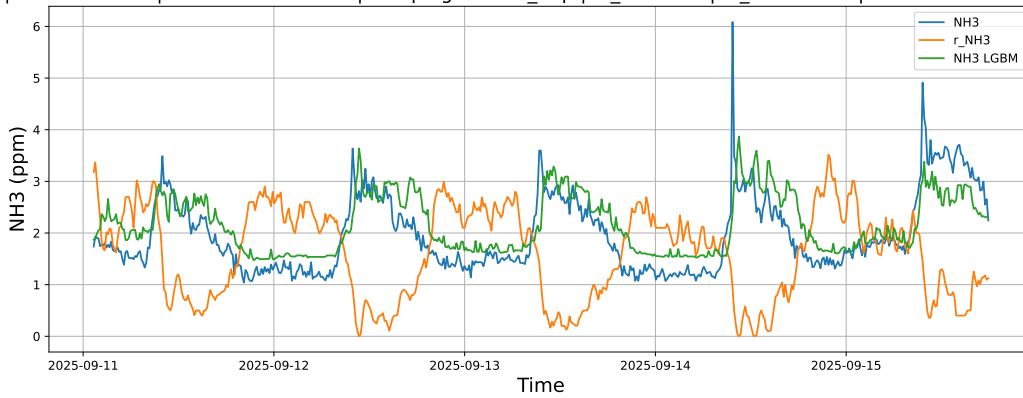


3.1.8.3 Time window : ALL

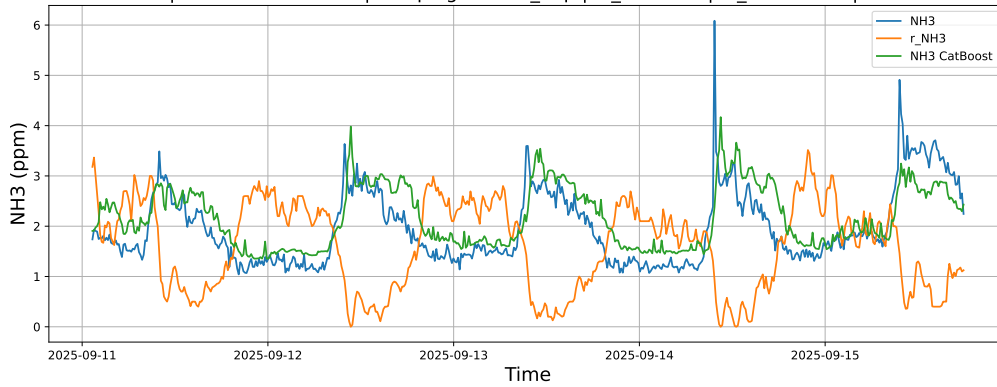
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

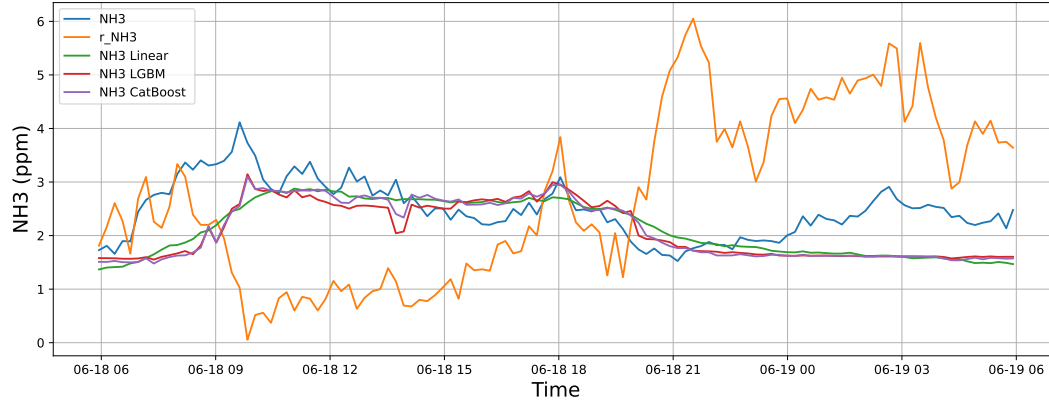


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

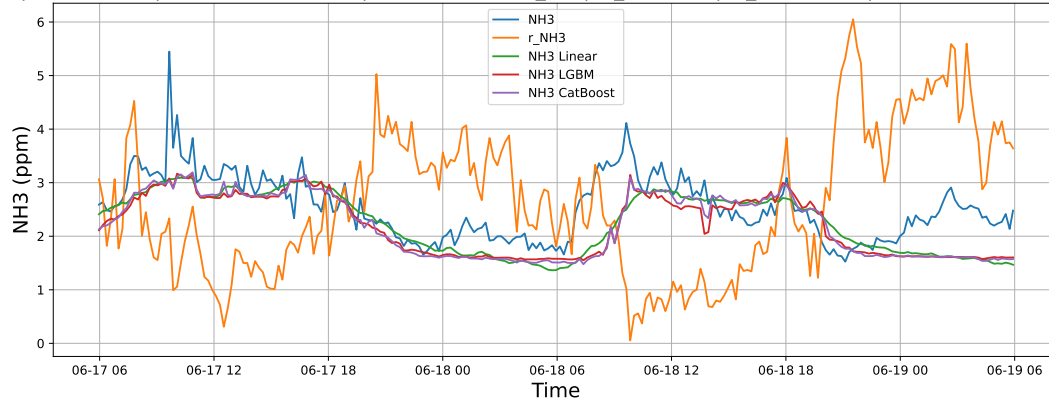
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



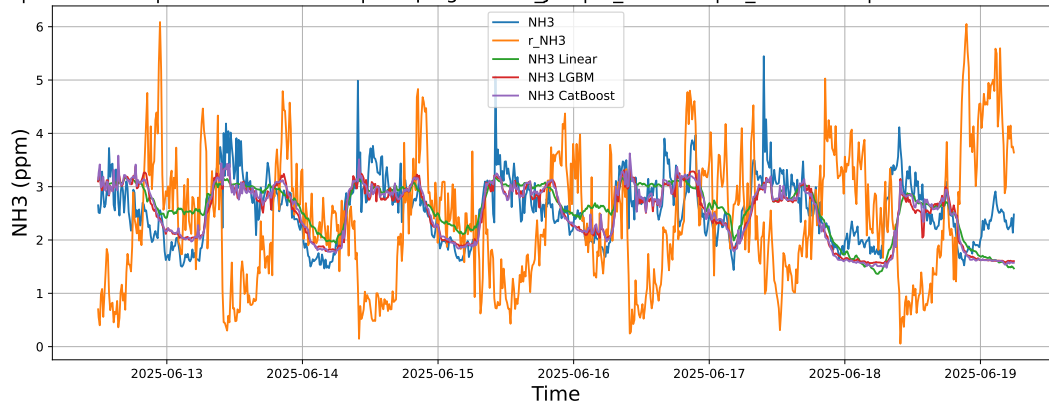
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

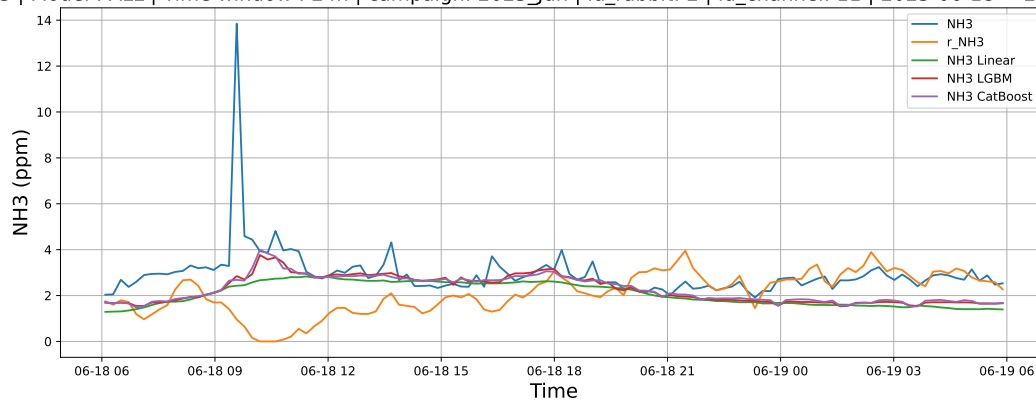
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

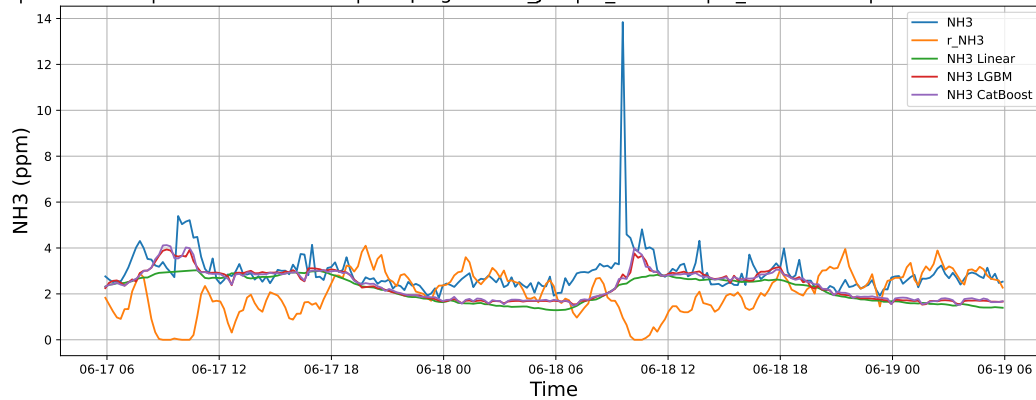
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



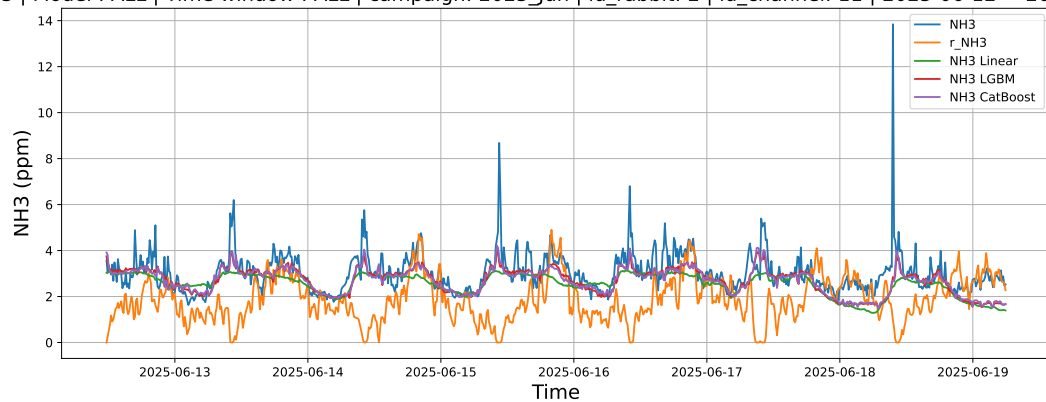
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

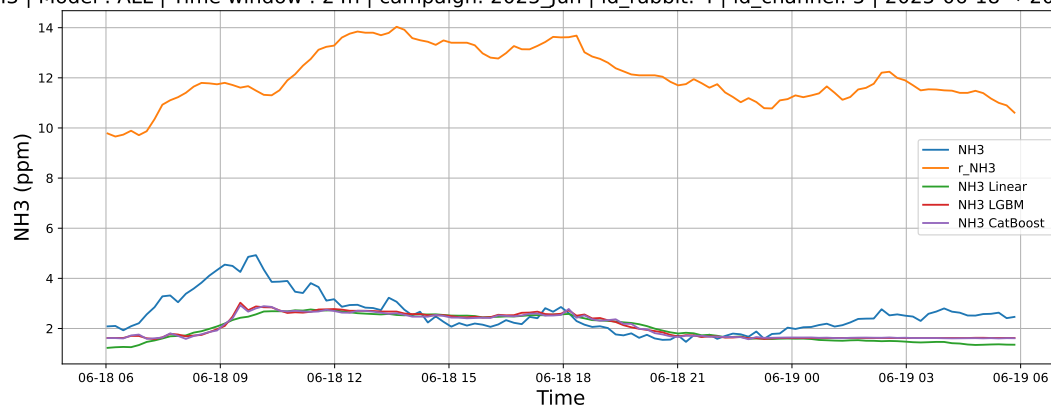
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

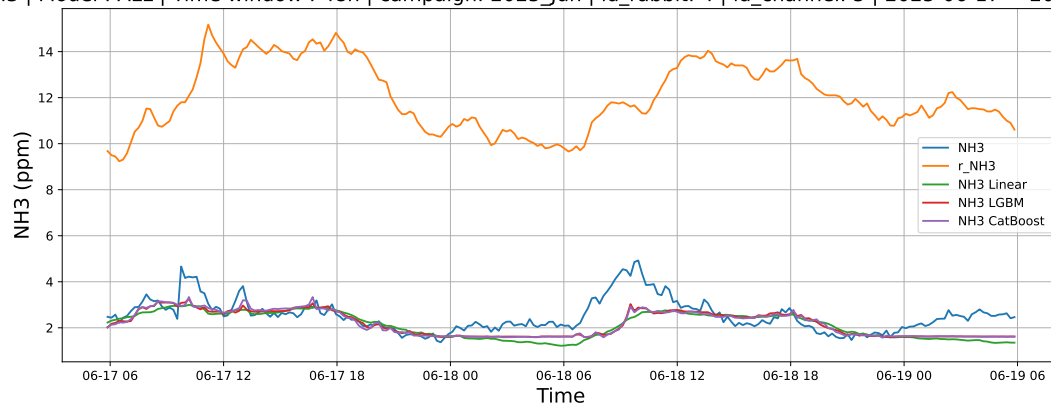
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



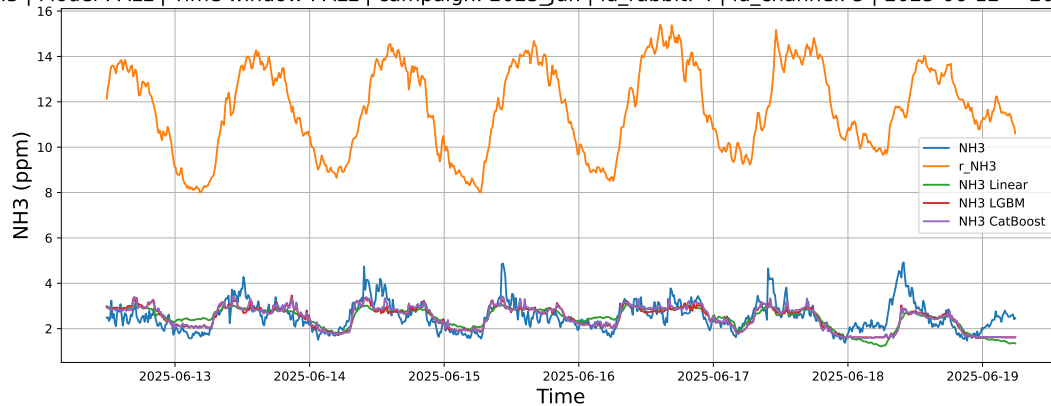
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

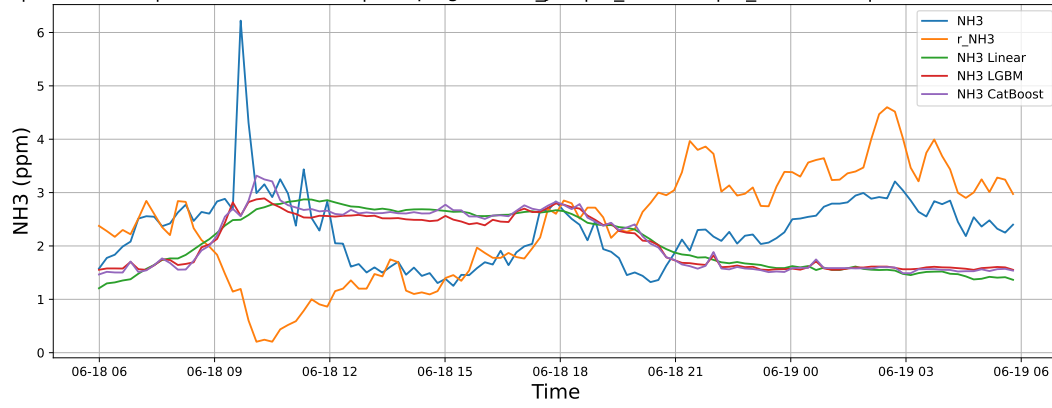
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

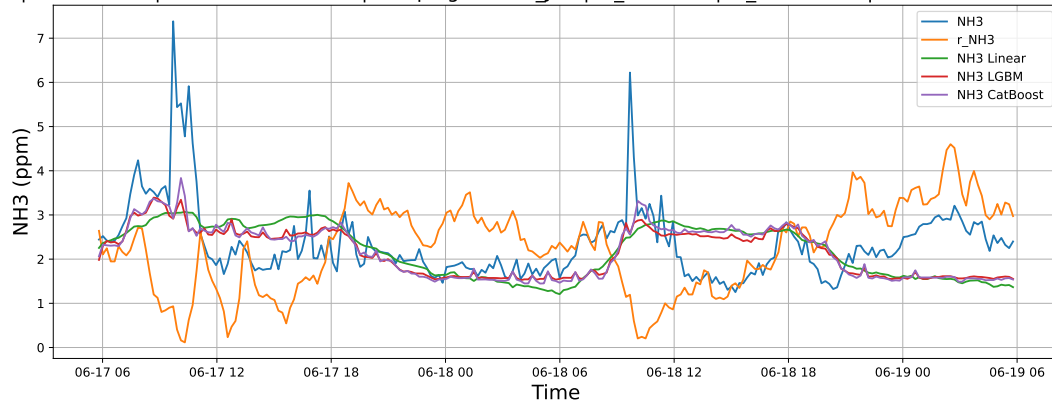
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



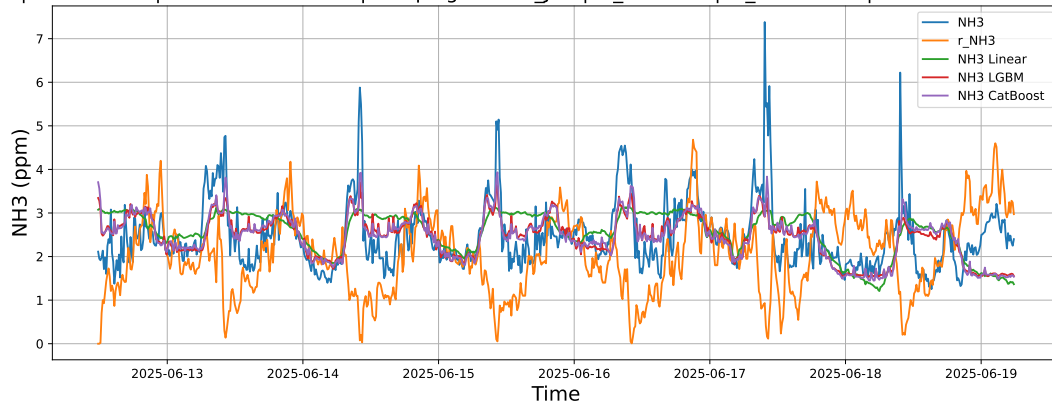
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

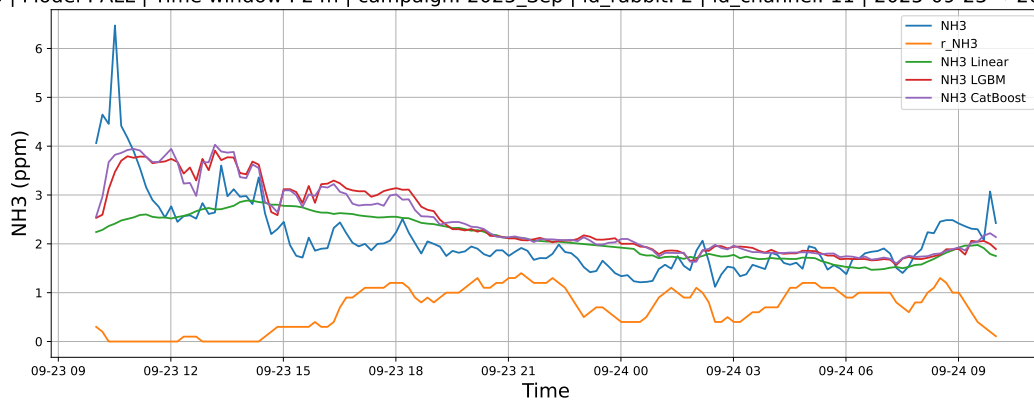
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

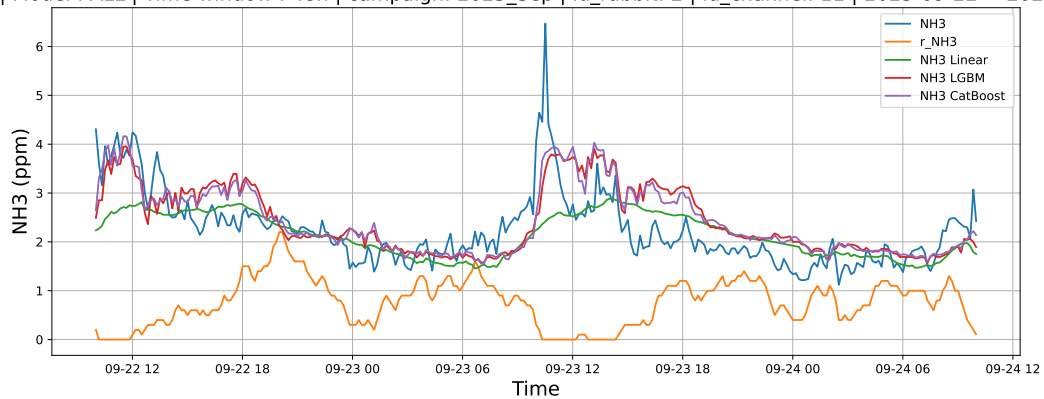
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



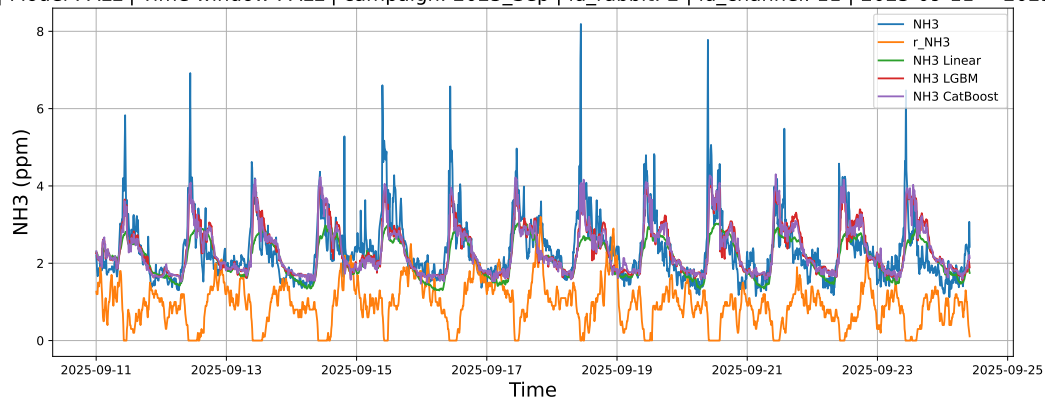
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

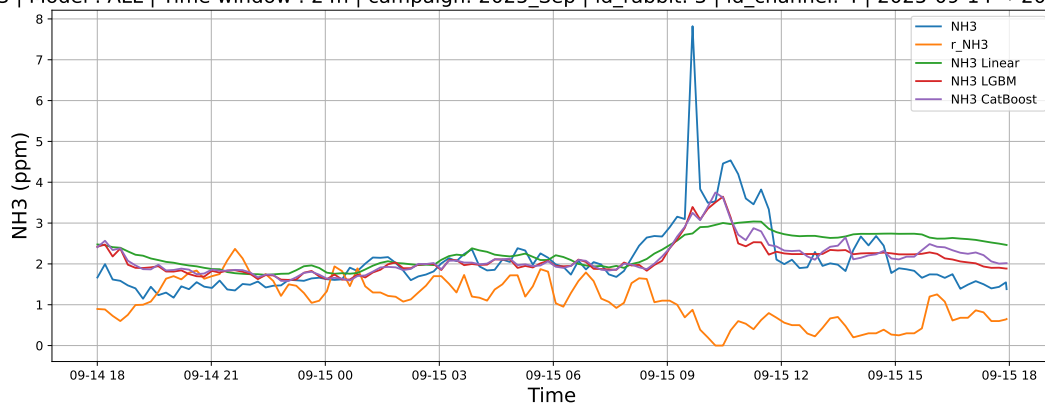
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

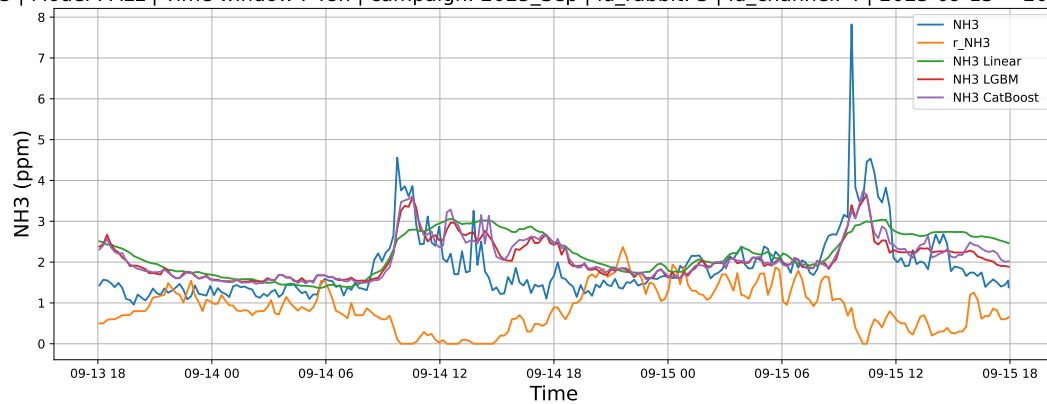
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



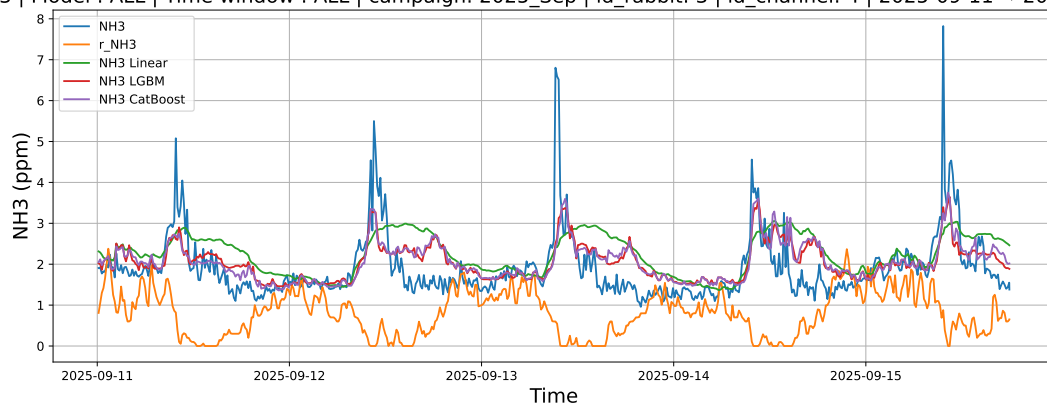
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

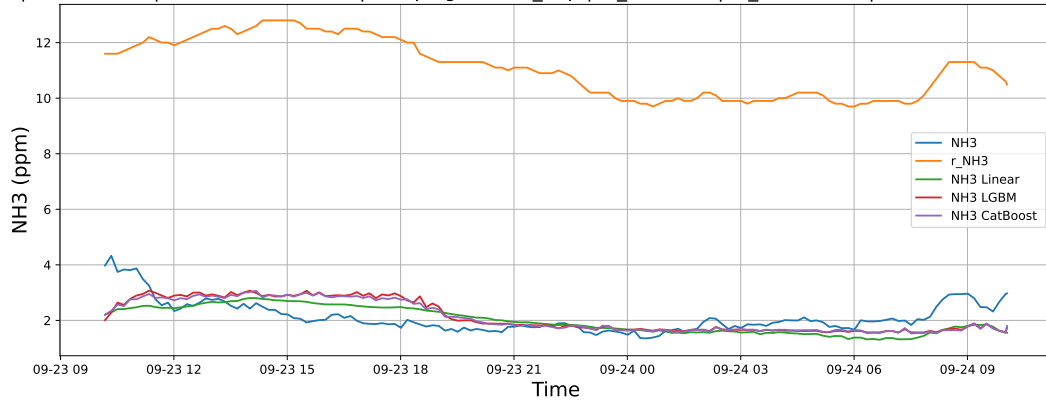
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

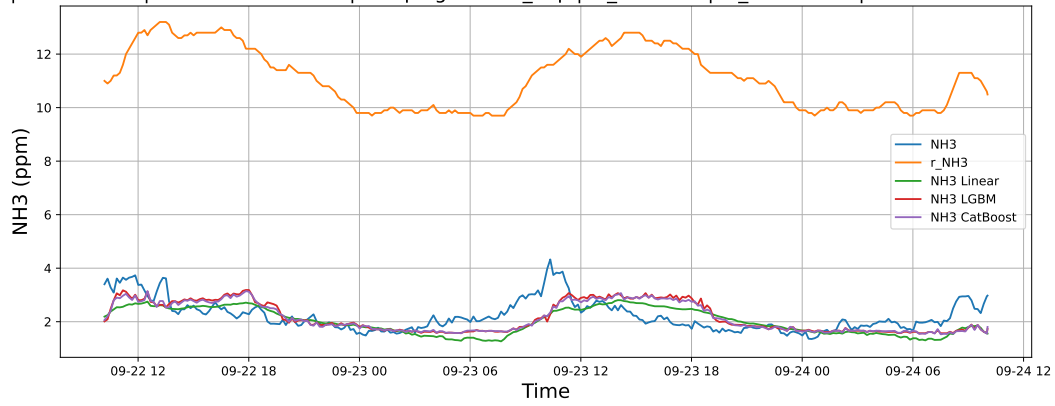
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



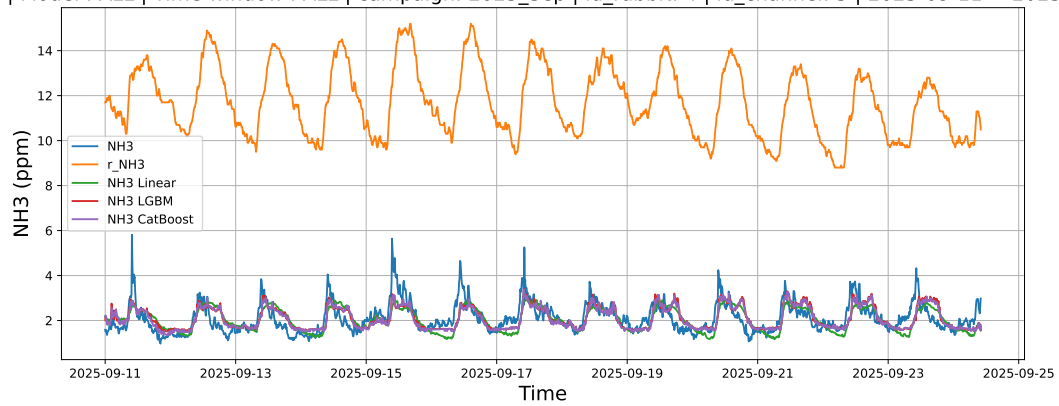
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

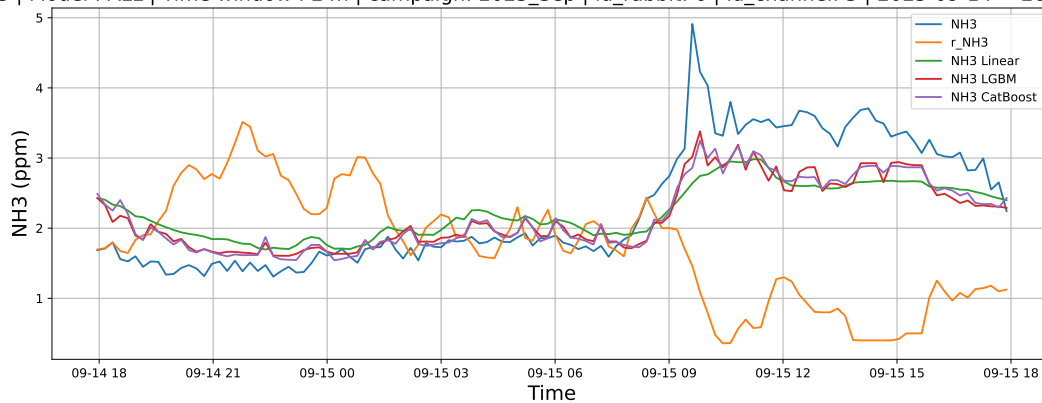
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

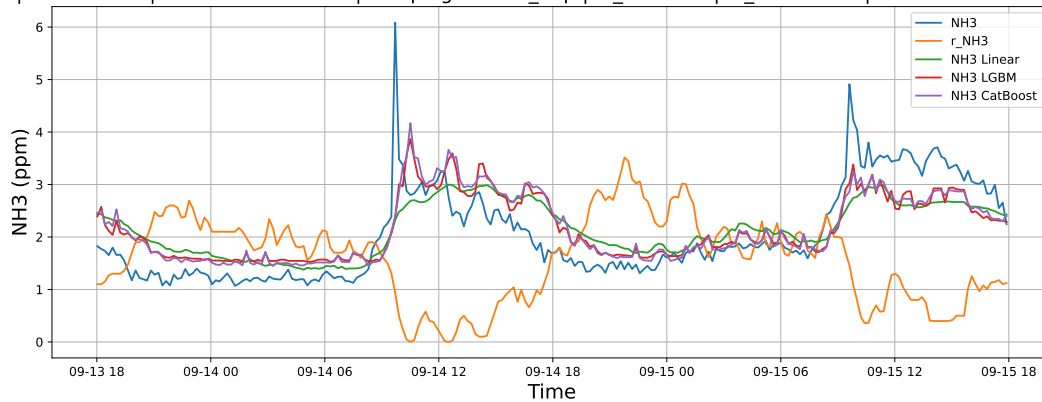
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

